



Leaky LWE: Learning with Errors with Semi-Adaptive Secret- and Error-Leakage

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Abstract. The Learning with Errors (LWE) problem asks to distinguish noisy samples $\mathbf{s}^T \mathbf{A} + \mathbf{e}^T \bmod q$ from uniformly random values given the random matrix $\mathbf{A}$. In this work, we show that a variant called Leaky LWE, where the distinguisher receives additionally noisy leakages $(\mathbf{s}^T, \mathbf{e}^T) \cdot \mathbf{L} + \mathbf{f}^T$ of the LWE secret $\mathbf{s}$ and error $\mathbf{e}$ for low-norm matrix $\mathbf{L}$ chosen adaptively by the distinguisher after seeing $\mathbf{A}$, is not easier than the standard LWE of the same dimensions up to polynomial losses in the noise level and the modulus. More generally, we show that the Leaky LWE problem is hard even if the public matrix $\mathbf{A}$ is structured and/or hinted and if the non-leaky parts of the secret and error do not follow Gaussian distributions, as long as the corresponding LWE problem without leakage is hard.

Our reduction from LWE to Leaky LWE unifies and extends prior results on the Error-Leakage LWE problem [Döttling-Kolonelos-Lai-Lin-Malavolta-Rahimi, EUROCRYPT'23], where $\mathbf{L}$ only acts on the error $\mathbf{e}$ and the Hint-MLWE problem [Kim-Lee-Seo-Song, CRYPTO'23], where $\mathbf{L}$ is restricted to concatenations of random Gaussian scalar matrices not controlled by the distinguisher. Previously, the Hint-MLWE and Error-Leakage LWE assumptions were used as computational replacements of the statistical noise flooding technique in security proofs which led to improved parameters in lattice-based cryptographic constructions such as zero-knowledge proofs, threshold signatures and registration-based encryption. We provide lemmas which abstract out such computational arguments based on Leaky LWE.

1 Introduction

The Learning with Errors (LWE) problem [Reg05] has been for two decades a cornerstone of lattice-based cryptography. It asks to distinguish samples of the form

$$\mathbf{b}^T = \mathbf{s}^T \mathbf{A} + \mathbf{e}^T \bmod q$$

from uniformly random $\mathbf{b} \leftarrow \mathbb{Z}_q^m$ given the matrix $\mathbf{A}$, where typically the matrix $\mathbf{A} \leftarrow \mathbb{Z}_q^{n \times m}$ is uniformly random, the secret $\mathbf{s} \in \mathbb{Z}_q^n$ is uniformly random or Gaussian and the low-norm error $\mathbf{e} \in \mathbb{Z}_q^m$ is Gaussian. The problem can be generalised to the so-called ring ($n = 1$) [SSTX09, LPR10] and module ($n > 1$) [BGV14, LS15] settings by replacing $\mathbb{Z}$ with an appropriate ring $\mathcal{R}$, typically the ring of integers of a cyclotomic field. All these settings are called LWE in this work.

1.1 LWE with Leakages

A recent growing trend in lattice-based cryptography is to study hardness of lattice problems in presence of additional structure and/or hints, which have been shown useful

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for constructing more functionally advanced and/or more efficient primitives. A particular family of such problems are LWE variants where the distinguisher further receives noisy leakages of the secret $\mathbf{s}$ and/or error $\mathbf{e}$ [DKL⁺23, KLSS23]. This family of problems is captured by experiments with the following template:

1. Sample a matrix $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ from some public distribution.
2. A leakage matrix $\mathbf{L} \in \mathcal{R}^{(n+m) \times Q}$ is chosen, either randomly from some public distribution or by the distinguisher after seeing the matrix $\mathbf{A}$.
3. Sample the LWE secret $\mathbf{s} \in \mathcal{R}_q^n$, LWE error $\mathbf{e} \in \mathcal{R}^m$ and leakage error $\mathbf{f} \in \mathcal{R}^Q$ from some public distributions. Let $\mathbf{b}$ be either an LWE sample $\mathbf{b}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \bmod q$ or sampled uniformly at random as $\mathbf{b} \leftarrow \mathcal{R}_q^m$. Let the leakage be $\mathbf{l}^\top = (\mathbf{s}^\top, \mathbf{e}^\top) \mathbf{L} + \mathbf{f}^\top$.
4. Given $(\mathbf{A}, \mathbf{b}, \mathbf{l})$, decide whether $\mathbf{b}$ is an LWE sample or uniformly random.

Both the recently proposed Error-Leakage LWE assumption of [DKL⁺23] and the Hint-MLWE assumption of [KLSS23] are special cases of the above template, with the leakage $\mathbf{l}$ being restricted to different special cases, and both have been shown useful for avoiding the need of the “noise flooding” technique¹ in security proofs, summarised below.

Leakages in security proofs. A recurring task in the security proofs of LWE-based constructions is as follows: Given $(\mathbf{A}, \hat{\mathbf{b}})$ where $\hat{\mathbf{b}}^\top = \hat{\mathbf{s}}^\top \mathbf{A} + \hat{\mathbf{e}}^\top \bmod q$, argue that tuples $(\mathbf{A}, \mathbf{b}, \mathbf{l})$ of the form

$$\mathbf{b}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \bmod q \quad \text{and} \quad \mathbf{l}^\top = (\mathbf{s}^\top, \mathbf{e}^\top) \mathbf{L} + \mathbf{f}^\top$$

are efficiently simulatable, where $\mathbf{s}, \hat{\mathbf{s}}, \mathbf{e}, \hat{\mathbf{e}}, \mathbf{f}$ are sampled from appropriate distributions. Furthermore, if $(\mathbf{A}, \hat{\mathbf{b}})$ is pseudorandom, then ideally $(\mathbf{A}, \mathbf{b})$ also is, even in presence of the leakage $\mathbf{l}$.

LWE with the above form of leakages commonly appears in security proofs of various types of cryptographic primitives, both explicitly or implicitly. For example, in a typical lattice-based Σ -protocol, $(\mathbf{A}, \mathbf{b})$ might play the role of a commitment or a public key, while $\mathbf{l}$ represents the prover’s response [KLSS23]. Argument of this type may appear when proving the zero-knowledge property of the Σ -protocol. Another class of example is the dual-Regev-style encryption schemes, where the ciphertext consists of

$$(\mathbf{b}^\top, \mathbf{c}^\top) = (\mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top, \mathbf{s}^\top \mathbf{A} \mathbf{L} - \mathbf{f}^\top) \bmod q$$

for some low-norm matrix $\mathbf{L}$. Such ciphertext can be simulated upon given

$$\mathbf{b}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \bmod q \quad \text{and} \quad \mathbf{l}^\top = \mathbf{e}^\top \mathbf{L} + \mathbf{f}^\top,$$

by expressing $\mathbf{c}$ as $\mathbf{c}^\top = \mathbf{b}^\top \mathbf{L} - \mathbf{l}^\top \bmod q$. Suppose $\mathbf{b}$ is pseudorandom even in presense of $\mathbf{l}$, then $(\mathbf{b}, \mathbf{c})$ can further be shown jointly pseudorandom, by either the leftover hash lemma or (another) invocation of LWE, hence proving semantic security of the encryption scheme.

Noise flooding and sidestepping it. To argue that $(\mathbf{A}, \mathbf{b}, \mathbf{l})$ is efficiently simulatable from $(\mathbf{A}, \hat{\mathbf{b}})$, a well-established technique is “noise flooding”: If the Gaussian width of $\mathbf{f}$ is super-polynomially (in the security parameter λ) greater than $\|(\mathbf{s}^\top, \mathbf{e}^\top) \mathbf{L}\|$, then the distributions of $\mathbf{f}$ and $\mathbf{l}$ are statistically close (in λ). This allows to decouple the distribution of $\mathbf{l}$ from that of $(\mathbf{A}, \mathbf{b})$, making them trivially simulatable given $(\mathbf{A}, \hat{\mathbf{b}})$. A major limitation of such statistical noise flooding argument is that, because of the lower bound of the Gaussian width of $\mathbf{f}$, the modulus q is forced to be super-polynomially large. For the LWE

¹Also called “noise-drowning” or “noise-smudging”.

problem to remain hard, one then needs to pick a large secret dimension n , which leads to unfavourable performance of the cryptographic construction.

As realised in [DKL⁺23, KLSS23], an alternative approach is to interpret and tolerate $\mathbf{I}^T = (\mathbf{s}^T, \mathbf{e}^T)\mathbf{L} + \mathbf{f}^T$ as noisy leakages of $(\mathbf{s}, \mathbf{e})$. If the LWE instance $(\mathbf{A}, \mathbf{b})$ remains pseudorandom even given such leakages, then the above security argument can still go through without resorting to statistical noise flooding. To achieve so, [DKL⁺23] formulated the Error-Leakage LWE assumption, where the leakage matrix $\mathbf{L}$ (only) acts on the LWE error $\mathbf{e}$, and which was used in arguing security of a dual-Regev-style encryption scheme (i.e. the second example in the above). [KLSS23] formulated the Hint-MLWE assumption, where the leakage matrix $\mathbf{L}$ is restricted to the form $\mathbf{L} = \mathbf{r}^T \otimes \mathbf{I}_{n+m}$ with $\mathbf{r} \in \mathcal{R}^Q$ sampled from some Gaussian distribution (not chosen by the distinguisher), and which was used for arguing zero-knowledge of a Σ -protocol (i.e. the first example in the above).

Both of these problems were shown to be as hard as standard LWE with appropriate parameters [DKL⁺23, KLSS23], ultimately allowing improved parameters while still basing security on standard LWE. Specifically, these assumptions have enabled a modulus q which is proportional to $\sqrt{Q}$, where $Q \in \text{poly}(\lambda)$ is the number of columns of $\mathbf{L}$ typically representing the number of queries that the adversary makes to some oracles [DKL⁺23, KLSS23, DKM⁺24, BKL⁺25]. Typically, Q is set to e.g. 2^{60} , translating to a 30-bit increase of the modulus q , which is more practical than e.g. setting $q > 2^{128}$.

Despite the demonstrated applications, the Error-Leakage LWE [DKL⁺23] and Hint-MLWE [KLSS23] assumptions are limited in different ways: The former is unable to support any leakage on the LWE secret, whereas the latter is tailored to support leakage matrix $\mathbf{L}$ of a very specific algebraic structure and distribution. For applications which demand more flexibility, both assumptions fall short. For example, when attempting to avoid noise flooding in Σ -protocols where the challenges are not Gaussian and/or the responses take a different algebraic structure, this translates to LWE which supports leakages on both the secret $\mathbf{s}$ and error $\mathbf{e}$ under a leakage matrix $\mathbf{L}$ of other desired structures, which is not supported by any of the prior two assumptions. Another unsupported scenario is to reason about hardness of LWE when given additional LWE samples $\mathbf{s}^T\mathbf{B} + \mathbf{f}^T$ whose matrix $\mathbf{B}$ is potentially correlated with $\mathbf{A}$, e.g. variants of the Reused-A LWE assumption of [MS24].²

1.2 Our Contribution

We present the Leaky LWE assumption, which captures the above “LWE with leakages” template in a general way and whose flexibility allows to fill the mentioned gap. Specifically, this work provides the following:

1. We formalise the Leaky LWE assumption, whose interface supports the following:
 - Leakages on both the LWE secret $\mathbf{s}$ and error $\mathbf{e}$.
 - Leakage matrix $\mathbf{L}$ which can be semi-adaptively chosen by the adversary, i.e. after seeing the public matrix $\mathbf{A}$, but before seeing the LWE sample $\mathbf{s}^T\mathbf{A} + \mathbf{e}^T \bmod q$.
 - LWE samples $\mathbf{s}^T\mathbf{A} + \mathbf{e}^T \bmod q$ where the matrix $\mathbf{A}$ sampled as $(\mathbf{A}, \mathbf{O}) \leftarrow D_{\mathbf{A}}$ can be arbitrarily structured and possibly come with a correlated oracle $\mathbf{O}$.

Leaky LWE can be seen as generalisations and unification of the Error-Leakage LWE and Hint-MLWE assumptions, achieving the best of both worlds.³

²The Reused-A LWE of [MS24] is not captured by the above “LWE with leakages” template, since the corresponding leakage is not bounded-norm. The template however captures assumptions in the same flavour up to the bounded-norm constraint.

³Technically, since Hint-MLWE is weaker in the sense that the leakage matrix is chosen at random, we recover an assumption which implies the Hint-MLWE assumption.

2. Generalising the reductions of [DKL⁺23, KLSS23], we show that Leaky LWE for any distribution D_A is implied by LWE with respect to the same distribution D_A with appropriate parameters. In particular, the “standard” variant of Leaky LWE with uniform matrices (and no oracle) is implied by standard LWE. Our reduction for Leaky LWE is similar to that in [DKL⁺23, KLSS23] for Error-Leakage LWE and Hint-MLWE respectively, but with several differences in the details. Compared to [KLSS23], our reduction handles unstructured leakage matrices chosen by the adversary, arbitrary rings and any of their embeddings, and non-spherical Gaussian distributions. Compared to [DKL⁺23], we directly analyse the concerned discrete Gaussian distributions without taking the detour of [DKL⁺23] to the continuous Gaussian distributions, which results in cleaner theorem statements, slightly wider range of supported parameters, and significantly simpler proofs. In passing, we also identify multiple gaps and errors in the statements and proofs of [DKL⁺23]. We detail the issues and discuss the corresponding fixes in Section A.
3. We provide utility lemmas which abstract out the “computational noise flooding” argument. These lemmas serve dual purposes: They can be directly plugged into typical security proofs because of the wide range of supported parameters, and their simple proofs from Leaky LWE (therefore also from LWE) demonstrate how the interface of Leaky LWE can be flexibly utilised.

1.3 Technical Overview

We give an overview of our reduction from LWE to Leaky LWE. To recall again, a Leaky LWE instance consists of (1) an LWE matrix $\mathbf{A}$, (2) an LWE challenge $\mathbf{b}^T$, which either equals $\mathbf{s}^T \mathbf{A} + \mathbf{e}^T \bmod q$ for some secret $\mathbf{s}$ and error $\mathbf{e}$, or is uniformly random, and (3) some leakage $(\mathbf{s}^T, \mathbf{e}^T) \mathbf{L} + \mathbf{f}^T$, where $\mathbf{L}$ is an arbitrary leakage matrix chosen by an adversary after seeing $\mathbf{A}$ (potentially subject to constraints which are part of the assumption parameters). The reduction has three main steps, as follows:

1. We observe that when the LWE secret and error $\mathbf{x}^T := (\mathbf{s}^T, \mathbf{e}^T)$ follow the Gaussian distribution, then the leakage $\mathbf{x}^T \cdot \mathbf{L} + \mathbf{f}^T$, which is a noisy linear function in $\mathbf{x}^T$, also follows Gaussian, if $\mathbf{f}$ is Gaussian. By appealing to lemmas on Gaussian linear transforms, one can show that $(\mathbf{x}, \mathbf{x}^T \cdot \mathbf{L} + \mathbf{f}^T) \equiv (\mathbf{y}_0, \mathbf{y}_1^T)$, where $\mathbf{y}_0, \mathbf{y}_1$ are sampled in reverse order: First sample $\mathbf{y}_1$ from some Gaussian with appropriate parameters (dependent on $\mathbf{L}$), then sample $\mathbf{y}_0$ from a correlated Gaussian distribution (dependent on both $\mathbf{L}$ and $\mathbf{y}_1$).
2. We decompose the Gaussian sample $\mathbf{y}_0$ into a sum of two Gaussians, $\mathbf{y}_0 = \mathbf{y}_{00} + \mathbf{y}_{01}$, which is possible thanks again to the rich properties of Gaussians. In particular, although $\mathbf{y}_0$ depends on $\mathbf{L}$, it is possible to set parameters such that one of the two summands, say $\mathbf{y}_{00}$, is independent of $\mathbf{L}$. Overall, we obtain

$$(\mathbf{x}, \mathbf{x}^T \cdot \mathbf{L} + \mathbf{f}^T) \approx_s (\mathbf{y}_{00} + \mathbf{y}_{01}, \mathbf{y}_1^T).$$

3. We construct a reduction from LWE. Given an LWE instance $(\mathbf{A}, \mathbf{b})$, the reduction samples $(\mathbf{y}_{01}, \mathbf{y}_1)$ according to parameters derived in the above two steps, call $\mathbf{y}_{01}^T = (\hat{\mathbf{s}}^T, \hat{\mathbf{e}}^T)$, and return $(\mathbf{A}, \mathbf{b} + \hat{\mathbf{s}}^T \mathbf{A} + \hat{\mathbf{e}}^T \bmod q, \text{leakage} = \mathbf{y}_1)$. If the input is an LWE sample $\mathbf{b}^T = \mathbf{s}^T \mathbf{A} + \mathbf{e}^T \bmod q$, then by interpreting $(\hat{\mathbf{s}}^T, \hat{\mathbf{e}}^T) = \mathbf{y}_{00}$ in the second step, we immediately obtain that

$$\begin{aligned} (\mathbf{b} + \hat{\mathbf{s}}^T \mathbf{A} + \hat{\mathbf{e}}^T \bmod q, \mathbf{y}_1) &= ((\hat{\mathbf{s}}^T + \mathbf{s}^T) \mathbf{A} + (\hat{\mathbf{e}} + \mathbf{e}^T) \bmod q, \mathbf{y}_1) \\ &\approx_s (\mathbf{s}^T \mathbf{A} + \mathbf{e} \bmod q, (\mathbf{s}^T, \mathbf{e}^T) \mathbf{L} + \mathbf{f}), \end{aligned}$$

i.e. the reduction statistically simulates the desired LWE sample and leakage. Else, if the input is a uniform $\mathbf{b}$, then it is obvious that linear transform of a random vector is still random, and the simulation of the leakage follows the same argument as above.

The above three steps are formally captured by Theorems 1 to 3 in Sections 3 and 4.

If the LWE secret $\mathbf{s}$ is not Gaussian but the error $\mathbf{e}$ is, then we are able to use similar strategy to argue leakages only on the LWE error $\mathbf{e}$ (but not on $\mathbf{s}$). Note also that the above simulation is independent of the distribution of $\mathbf{A}$ (and any other external information on it), and as such the reduction captures also more advanced settings, e.g. when correlated oracles on $\mathbf{A}$ are given, as long as the underlying LWE instance is so.

At a high level, our reduction can be seen as a mixture of those of [DKL⁺23, KLSS23]: The first step above resembles [KLSS23], whereas the second follows [DKL⁺23].

1.4 Related Work

A related family of LWE variants consists of the Extended LWE [AP12], Multi-hint Extended LWE [ALS16] and MLWE with Linear Secret/Error Leakage [WLL24], which is similar to the Leaky LWE family in that both consider the pseudorandomness of LWE samples in presence of leakages on the LWE secret and/or error, and all of the three stated assumptions admit reductions from standard LWE. However, a major difference is that this family considers leakages which are exact rather than noisy (i.e. $\mathbf{f} = \mathbf{0}$), and in exchange the number of leakages (i.e. Q) is strictly smaller than the LWE secret dimension n . In contrast, the Leaky LWE family allows an arbitrarily large polynomial number Q of noisy leakages. For this difference, the techniques for obtaining reductions from LWE are quite different, and in this work we only focus on the Leaky LWE family.

In a concurrent and independent work, [BJTW25] studied (the search version of) LWE with leakages by analysing the residual entropy of an LWE error given the leakage. Since lower-bounding entropy is possible for a wider range of distributions, they are able to capture other leakage distributions, e.g. bounded support leakage and specific quadratic leakage. When focusing on noisy linear leakage (the setting that we concern), our results and theirs seem formally incomparable, both because of the difference between search- and decision-LWE and their extra reliance on the hardness of a Short Integer Solution (SIS) problem. Practically, we believe our results offer simpler interface for parameters estimation, due to the direct and dimension-preserving reduction from LWE, and the minimal conditions imposed. For example, for a (search-)-LWE problem of secret dimension n with leakages, the reduction of [BJTW25, Theorem 4.1] further requires a rank- $(m - n)$ SIS problem to be sufficiently hard, thus imposing a lower bound on m . Similar to our result, the entropy bound of [BJTW25, Section 6.1] shows a dependency of $\sigma_{\mathbf{f}}$ on $\sqrt{Q}$.

2 Preliminaries

We denote the security parameter by $\lambda \in \mathbb{N}$, and the sets of all polynomials and negligible functions in λ by $\text{poly}(\lambda)$ and $\text{negl}(\lambda)$ respectively. We write $\mathcal{X} \sim A$ to denote that $\mathcal{X}$ is a distribution over the support A . For discrete distributions $\mathcal{X}$ and $\mathcal{Y}$ over a countable set A , the statistical distance between $\mathcal{X}$ and $\mathcal{Y}$ is $\Delta(\mathcal{X}, \mathcal{Y}) = \frac{1}{2} \sum_{a \in A} |\mathcal{X}(a) - \mathcal{Y}(a)|$. Two ensembles of distributions $\mathcal{X} = \{\mathcal{X}_\lambda\}_{\lambda \in \mathbb{N}}$ and $\mathcal{Y} = \{\mathcal{Y}_\lambda\}_{\lambda \in \mathbb{N}}$ are said to be statistically close in λ if $\Delta(\mathcal{X}_\lambda, \mathcal{Y}_\lambda) \leq \text{negl}(\lambda)$, denoted also by $\mathcal{X} \approx_s \mathcal{Y}$. The ensembles $\mathcal{X}, \mathcal{Y}$ are said to be computationally close in λ , if for all probabilistic polynomial time (PPT) algorithms $\mathcal{A}$ and for any event $\mathcal{E}$, it holds that $|\Pr[\mathcal{A}(\mathcal{X}_\lambda) = \mathcal{E}] - \Pr[\mathcal{A}(\mathcal{Y}_\lambda) = \mathcal{E}]| \leq \text{negl}(\lambda)$, denoted also by $\mathcal{X} \approx_c \mathcal{Y}$.

For any $n \in \mathbb{N}$, we write $[n]$ for the set $\{1, \dots, n\}$. We use bold lower case letters (e.g. $\mathbf{x}$) to denote column vectors and bold upper case letters (e.g. $\mathbf{X}$) to denote matrices. For

a tuple of matrices $(\mathbf{A}_i)_i$, we write $\text{diag}((\mathbf{A}_i)_i)$ to denote the diagonal matrix with the matrices $\mathbf{A}_i$ along the main diagonal.

Algebraic number theory background. Throughout this work we let $\mathcal{K}$ be a number field that is a degree- φ extension of $\mathbb{Q}$, where we also call φ the degree of $\mathcal{K}$, and we denote by $\mathcal{R}$ a subring of $\mathcal{K}$. For example, one may think of $\mathcal{K} = \mathbb{Q}(\zeta)$ where $\zeta = \zeta_f \in \mathbb{C}$ is a root of the f -th cyclotomic polynomial of degree $\varphi = \varphi(f)$, and $\mathcal{R} = \mathbb{Z}[\zeta]$ its ring of integers. For $q \in \mathbb{N}$, define the quotient ring $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$.

We let $\psi : \mathcal{K} \rightarrow \mathbb{R}^\varphi$ be any $\mathbb{Q}$ -linear map which maps some integral basis $(z_1, \dots, z_\varphi)$ of $\mathcal{K}$ to a set of $\mathbb{R}$ -linearly independent vectors $(\psi(z_1), \dots, \psi(z_\varphi))$.⁴ We call such a map ψ an embedding of $\mathcal{K}$ to $\mathbb{R}^\varphi$. Common examples are the coefficient embedding and the canonical embedding of $\mathcal{K}$. For any embedding ψ there exists a unique $\mathbb{Q}$ -linear map $\Psi : \mathcal{K} \rightarrow \mathbb{R}^{\varphi \times \varphi}$ (dependent on ψ) such that for all $a, x \in \mathcal{K}$ it holds that $\psi(a \cdot x) = \Psi(a) \cdot \psi(x)$. (See Lemma 1 for a proof.) The notation extends to vectors and matrices by applying the maps ψ and Ψ element-wise, e.g. for a vector $\mathbf{a} = (a_1, \dots, a_n) \in \mathcal{K}^n$, we write $\psi(\mathbf{a})^\top = (\psi(a_1)^\top, \dots, \psi(a_n)^\top)$. With this extension we observe that for all $\mathbf{A} \in \mathcal{K}^{n \times m}$, $\mathbf{x} \in \mathcal{K}^m$, it holds that $\psi(\mathbf{A}\mathbf{x}) = \Psi(\mathbf{A}) \cdot \psi(\mathbf{x})$.

Lemma 1. *Let $\mathcal{K}$ be a number field of degree φ and ψ be an embedding of $\mathcal{K}$ to $\mathbb{R}^\varphi$ with respect to some integral basis $(z_1, \dots, z_\varphi)$ of $\mathcal{K}$. There exists a unique linear map $\Psi : \mathcal{K} \rightarrow \mathbb{R}^{\varphi \times \varphi}$ (dependent on ψ) such that for all $a, x \in \mathcal{K}$ it holds that $\psi(a \cdot x) = \Psi(a) \cdot \psi(x)$.*

Proof. Define the $\mathbb{Q}$ -linear map $\Gamma : \mathcal{K} \rightarrow \mathbb{R}^{\varphi \times \varphi}$ by

$$\Gamma(a) = (\psi(a \cdot z_1) \quad \dots \quad \psi(a \cdot z_\varphi))$$

from which we construct Ψ as $\Psi(a) = \Gamma(a) \cdot \Gamma(1)^{-1}$ for all $a \in \mathcal{K}$. Note that by hypothesis the columns of $\Gamma(1)$ are $\mathbb{R}$ -linearly independent. Therefore, $\Gamma(1)$ is invertible over $\mathbb{R}$ and Ψ is well-defined.

Next, we show that for any $a, x \in \mathcal{K}$ we have $\psi(a \cdot x) = \Psi(a) \cdot \psi(x)$. Since $(z_1, \dots, z_\varphi)$ is an integral basis, for any $x \in \mathcal{K}$ we can express it as $x = \sum_{i \in [\varphi]} x_i \cdot z_i$ for some $x_i \in \mathbb{Q}$. Since ψ is linear, we have $\psi(x) = \sum_{i \in [\varphi]} x_i \cdot \psi(z_i)$ or equivalently $\psi(x) = \Gamma(1) \cdot (x_1, \dots, x_\varphi)^\top$. Observe that

$$\begin{aligned} \Psi(a) \cdot \psi(x) &= \Gamma(a) \cdot \Gamma(1)^{-1} \cdot \Gamma(1) \cdot (x_1, \dots, x_\varphi)^\top \\ &= (\psi(a \cdot z_1) \quad \dots \quad \psi(a \cdot z_\varphi)) \cdot (x_1, \dots, x_\varphi)^\top \\ &= \sum_{i \in [\varphi]} x_i \cdot \psi(a \cdot z_i) = \psi(a \cdot x) \end{aligned}$$

as desired, where for the third equality we used the linearity of ψ .

Finally, we show the uniqueness of Ψ . Suppose there exists two maps Ψ_0, Ψ_1 satisfying $\psi(a \cdot x) = \Psi_i(a) \cdot \psi(x)$ any $a, x \in \mathcal{K}$. Substituting $x = z_i$ for all $i \in [\varphi]$, we have

$$\begin{aligned} \Psi_i(a) \cdot \Gamma(1) &= \Gamma(a), \\ (\Psi_0(a) - \Psi_1(a)) \cdot \Gamma(1) &= \mathbf{0}. \end{aligned}$$

Since $\Gamma(1)$ is invertible, we conclude that $\Psi_0(a) = \Psi_1(a)$ for all $a \in \mathcal{K}$, i.e. $\Psi_0 = \Psi_1$. $\square$

For example, when taking ψ to be the canonical embedding (see e.g. [LPR13]), then we have

$$\Psi(\mathbf{A}) = \begin{pmatrix} \text{diag}(\psi(a_{11})) & \dots & \text{diag}(\psi(a_{1m})) \\ \vdots & & \vdots \\ \text{diag}(\psi(a_{n1})) & \dots & \text{diag}(\psi(a_{nm})) \end{pmatrix}.$$

⁴It can be shown that if ψ maps some integral basis of $\mathcal{K}$ to a set of φ $\mathbb{R}$ -linearly independent vectors then it does so for all integral bases of $\mathcal{K}$. We omit the proof because we do not need this property.

When taking $\mathcal{K}$ to be the f -th cyclotomic field with f a power of 2 and ψ to be the coefficient embedding with respect to the standard power basis (see e.g. [LPR13]), then we have

$$\Psi(\mathbf{A}) = \begin{pmatrix} \text{rot}(a_{11}) & \dots & \text{rot}(a_{1m}) \\ \vdots & & \vdots \\ \text{rot}(a_{n1}) & \dots & \text{rot}(a_{nm}) \end{pmatrix}, \text{ where } \text{rot}(x) = \begin{pmatrix} x_1 & -x_\varphi & \dots & -x_2 \\ x_2 & x_1 & & -x_3 \\ \vdots & & & \vdots \\ x_\varphi & x_{\varphi-1} & \dots & x_1 \end{pmatrix}$$

for any $x \in \mathcal{K}$ with embedding $\psi(x) = (x_1, \dots, x_\varphi)^\top$.

Norms. For a vector $\mathbf{a} = (a_1, \dots, a_n)^\top \in \mathbb{R}^n$, its ℓ_2 -norm is $\|\mathbf{a}\| := (\sum_{i \in [n]} a_i^2)^{1/2}$. For an element $a \in \mathcal{K}$, its norm (w.r.t. the embedding ψ) is $\|a\| := \|\psi(a)\|$. For a vector $\mathbf{a} = (a_1, \dots, a_n) \in \mathcal{K}^n$, its norm (w.r.t. ψ) is $\|\mathbf{a}\| := \|(\psi(a_1), \dots, \psi(a_n))^\top\|$. For a matrix $\mathbf{A} \in \mathcal{K}^{n \times m}$, its spectral norm (w.r.t. ψ) is

$$\|\mathbf{A}\| := \sup_{\mathbf{x} \in \mathcal{K}^m} \frac{\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|}.$$

Singular values. For a matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$, its i -th singular value is the square root of the (necessarily non-negative) i -th largest eigenvalue of $\mathbf{A}^\top \mathbf{A}$. Its maximum (i.e. the 1-st) singular value is denoted by $s_{\max}(\mathbf{A})$, and its minimum (i.e. the $\max(n, m)$ -th) singular value is denoted by $s_{\min}(\mathbf{A})$. When writing $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ in its singular decomposition, where $\mathbf{U} \in \mathbb{R}^{n \times n}$, $\mathbf{V} \in \mathbb{R}^{m \times m}$ are orthogonal matrices and $\mathbf{D} \in \mathbb{R}^{n \times m}$ a rectangular diagonal matrix with (necessarily non-negative) diagonal entries. The i -th largest diagonal entry of $\mathbf{D}$ is the i -th singular value of $\mathbf{A}$.

For a matrix $\mathbf{A} \in \mathcal{K}^{n \times m}$, its maximum and minimum singular values (w.r.t. ψ) are defined as

$$s_{\max}(\mathbf{A}) := s_{\max}(\Psi(\mathbf{A})) \quad \text{and} \quad s_{\min}(\mathbf{A}) := s_{\min}(\Psi(\mathbf{A})).$$

The following lemma relates the maximum singular value of a matrix to its norm over any embedding.

Lemma 2. *Let $\mathcal{K}$ be a number field with embedding ψ to $\mathbb{R}^\varphi$ and let $\mathbf{A} \in \mathcal{K}^{n \times m}$. It holds that $\|\mathbf{A}\| = s_{\max}(\mathbf{A})$.*

Proof. By the definition of $\|\mathbf{A}\|$ and $s_{\max}(\mathbf{A})$ and by Lemma 1, we have

$$\begin{aligned} \|\mathbf{A}\| &= \sup_{\mathbf{x} \in \mathcal{K}^m} \frac{\|\psi(\mathbf{A}\mathbf{x})\|}{\|\psi(\mathbf{x})\|} = \sup_{\mathbf{x} \in \mathcal{K}^m} \frac{\|\Psi(\mathbf{A}) \cdot \psi(\mathbf{x})\|}{\|\psi(\mathbf{x})\|} \\ &= \sup_{\mathbf{y} \in \mathbb{R}^{\varphi m}} \frac{\|\Psi(\mathbf{A}) \cdot \mathbf{y}\|}{\|\mathbf{y}\|} = s_{\max}(\Psi(\mathbf{A})) = s_{\max}(\mathbf{A}) \end{aligned}$$

where for the third equality we used the well-known fact that for finite-dimensional $\mathbb{R}$ -matrices the maximum singular value and the spectral norm coincide, and for the fourth we observe that $\frac{\|\Psi(\mathbf{A}) \cdot \psi(\mathbf{x})\|}{\|\psi(\mathbf{x})\|}$ for $\mathbf{x} \in \mathcal{K}^m$ can be infinitely close to $\sup_{\mathbf{y} \in \mathbb{R}^{\varphi m}} \frac{\|\Psi(\mathbf{A}) \cdot \mathbf{y}\|}{\|\mathbf{y}\|}$. $\square$

Positive definite matrices. A symmetric matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ is said to be positive definite (resp. positive semi-definite), denoted by $\mathbf{M} > 0$ (resp. $\mathbf{M} \geq 0$), if $\mathbf{x}^\top \mathbf{M} \mathbf{x}$ is positive (resp. non-negative) for all non-zero $\mathbf{x} \in \mathbb{R}^n$. For two symmetric matrices $\mathbf{M}, \mathbf{N}$, the notation $\mathbf{M} > \mathbf{N}$ refers to $\mathbf{M} - \mathbf{N} > 0$, i.e. $\mathbf{M} - \mathbf{N}$ is positive definite. For any $c \in \mathbb{R}$, we also write $\mathbf{M} > c$ to denote $\mathbf{M} > c \cdot \mathbf{I}_n$. The notation $\mathbf{M} \geq \mathbf{N}$ and $\mathbf{M} \geq c$ are analogous. The following facts are elementary:

- A symmetric matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ is positive semi-definite if and only if it can be written as $\mathbf{M} = \mathbf{B}\mathbf{B}^\top$ for some $\mathbf{B} \in \mathbb{R}^{n \times n}$. A positive semi-definite matrix is positive definite if and only if it is invertible. Every positive definite matrix is invertible.
- For a positive semi-definite matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$, it holds that $s_{\min}(\mathbf{M}) \cdot \mathbf{I}_n \leq \mathbf{M} \leq s_{\max}(\mathbf{M}) \cdot \mathbf{I}_n$. If $\mathbf{M}$ is positive definite, then also $s_{\max}(\mathbf{M}^{-1}) = s_{\min}(\mathbf{M})^{-1}$.
- For matrices $\mathbf{A}, \mathbf{B}, \mathbf{B}' \in \mathbb{R}^{n \times n}$, where $\mathbf{B}$ and $\mathbf{B}'$ are symmetric, if $\mathbf{B} \leq \mathbf{B}'$, then $\mathbf{A}\mathbf{B}\mathbf{A}^\top \leq \mathbf{A}\mathbf{B}'\mathbf{A}^\top$. The same holds with strict inequality if $\mathbf{A}$ is full-rank.

Because of the first fact, for a matrix $\Sigma \geq 0$, we can write $\sqrt{\Sigma}$ to refer to some $\mathbf{B}$ satisfying $\Sigma = \mathbf{B}\mathbf{B}^\top$ without ambiguity.

Lemma 3 ([HMT11, Proposition 8.3]). *We have $\|\mathbf{M}\| \leq \|\mathbf{A}\| + \|\mathbf{C}\|$ for any positive semi-definite matrix*

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{pmatrix}.$$

Lattices. An n -dimensional lattice Λ is a discrete additive subgroup of $\mathbb{R}^n$. A lattice $\Lambda \subseteq \mathbb{R}^n$ is said to be generated by a basis of $\mathbb{R}$ -linearly-independent vectors $\{\mathbf{b}_i\}_{i \in [d]} \subseteq (\mathbb{R}^n)^d$, if for $\mathbf{B} := (\mathbf{b}_1, \dots, \mathbf{b}_d)$, it holds that $\Lambda = \Lambda(\mathbf{B}) = \{\mathbf{x} \in \mathbb{R}^n : \exists \mathbf{u} \in \mathbb{Z}^d, \mathbf{B}\mathbf{u} = \mathbf{x}\}$. A lattice Λ is said to be full-rank if $d = m$.

Discrete Gaussians. For $\Sigma > 0$ and $\mathbf{c} \in \mathbb{R}^n$, the n -dimensional Gaussian function with parameter $\sqrt{\Sigma}$ and centre $\mathbf{c}$ is $\rho_{\sqrt{\Sigma}, \mathbf{c}}(\mathbf{x}) := \exp(-\pi \cdot (\mathbf{x} - \mathbf{c})^\top \cdot \Sigma^{-1} \cdot (\mathbf{x} - \mathbf{c}))$. The discrete Gaussian distribution over a lattice Λ with parameter $\sqrt{\Sigma}$ and centre $\mathbf{c}$ is defined as follows: For any $\mathbf{x} \in \Lambda$,

$$\mathcal{D}_{\Lambda, \sqrt{\Sigma}, \mathbf{c}}(\mathbf{x}) = \rho_{\sqrt{\Sigma}, \mathbf{c}}(\mathbf{x}) / \rho_{\sqrt{\Sigma}, \mathbf{c}}(\Lambda),$$

where $\rho_{\sqrt{\Sigma}, \mathbf{c}}(\Lambda) := \sum_{\mathbf{x} \in \Lambda} \rho_{\sqrt{\Sigma}, \mathbf{c}}(\mathbf{x})$ is a finite normalisation factor. When $\Sigma = \sigma^2 \cdot \mathbf{I}$, we simply write $\mathcal{D}_{\Lambda, \sigma, \mathbf{c}}$. When $\mathbf{c} = \mathbf{0}$, we omit the subscript $\mathbf{c}$.

Definition 1 (Smoothing parameter). For $\epsilon > 0$, the ϵ -smoothing parameter of a lattice Λ is defined as

$$\eta_\epsilon(\Lambda) := \inf\{s \in \mathbb{R}^+ : \rho_{1/s}(\Lambda^* \setminus \{\mathbf{0}\}) \leq \epsilon\},$$

where Λ^* is the dual lattice of Λ .

For a matrix $\mathbf{S}$, the expression $\mathbf{S} \geq \eta_\epsilon(\Lambda)$ refers to $\mathbf{S} \geq \eta_\epsilon(\Lambda) \cdot \mathbf{I}$.⁵ We will rely on the following lemmas which concern linear transforms on discrete Gaussians over lattices.

Lemma 4 ([GMPW20, Lemma 1 (eprint Lemma 2.3)]). *For any $A \subseteq \mathbb{R}^n$ and matrices $\mathbf{S}, \mathbf{T}$ representing linear functions where $\mathbf{T}$ is injective on A , we have $\mathbf{T} \cdot \mathcal{D}_{A, \mathbf{S}} = \mathcal{D}_{\mathbf{T}A, \mathbf{T}\mathbf{S}}$.*

Lemma 5 (Adapted from [GMPW20, Theorem 4 (eprint Theorem 4.6)]). *Let $\epsilon \in (0, 1)$, and let $\mathbf{z} = (z_1, \dots, z_m)^\top \in \mathbb{Z}^m \setminus \{\mathbf{0}\}$. For $i \in [m]$, let $\Lambda_i + \mathbf{c}_i \subset \mathbb{R}^n$ be lattice cosets such that $\Lambda_\cap = \cap_i \Lambda_i$ is full-rank, and let $\mathbf{S}_i \in \mathbb{R}^{n \times n}$. If*

$$\text{diag}(\mathbf{S}_1, \dots, \mathbf{S}_m) \geq \eta_\epsilon(\ker(\mathbf{z}^\top \otimes \mathbf{I}_n) \cap \Lambda) \quad (1)$$

where $\Lambda = \Lambda_1 \times \dots \times \Lambda_m$, then

$$\Delta\left(\sum_{i \in [m]} z_i \cdot \mathcal{D}_{\Lambda_i + \mathbf{c}_i, \mathbf{S}_i}, \mathcal{D}_{\Lambda', \mathbf{S}'}\right) \leq \frac{2\epsilon}{1-\epsilon},$$

where $\Lambda' = \sum_{i \in [m]} z_i \cdot (\Lambda_i + \mathbf{c}_i)$ and $\mathbf{S}' = \sqrt{\sum_{i \in [m]} z_i \cdot \mathbf{S}_i \mathbf{S}_i^\top}$.

⁵This is consistent with e.g. the definition of smoothing parameter of [MP12], which $\mathbf{S} \geq \eta_\epsilon(\Lambda)$ is used to refer to $\rho_{\mathbf{S}, \mathbf{0}}(\Lambda^*) \leq 1 + \epsilon$.

The following lemma is implicit in the proof of [GMPW20, Theorem 4 (eprint Theorem 4.6)] (i.e. Lemma 5 quoted above).

Lemma 6. *Let $\epsilon \in (0, 1)$, and let $\mathbf{z} = (z_1, \dots, z_m)^\top \in \mathbb{Z}^m \setminus \{0\}$. For $i \in [m]$, let $\Lambda_i + \mathbf{c}_i \subset \mathbb{R}^n$ be lattice cosets such that $\Lambda_\cap = \cap_i \Lambda_i$ is full-rank, and let $\mathbf{S}_i \in \mathbb{R}^{n \times n}$.*

If for each $i \in [m]$ it holds that $\mathbf{S}_i = s_i \cdot \mathbf{I}_n$ for some $s_i > 0$, then Eq. (1) is implied by

$$((z_{i^*}/s_{i^*})^2 + \max_{i \neq i^*} (z_i/s_i)^2)^{-1/2} \geq \eta_\epsilon(\Lambda_\cap),$$

where i^* minimises $|z_{i^*}/s_{i^*}| \neq 0$.

The following is a simple consequence of Lemma 6.

Lemma 7. *Let $\Lambda \in \mathbb{R}^n$ be a full-rank lattice, let $\mathbf{c} \in \mathbb{R}^n$, and let $\Sigma_1, \Sigma_2 \in \mathbb{R}^{n \times n}$ be positive definite matrices. If $(s_{\min}(\Sigma_1)^{-1} + s_{\min}(\Sigma_2)^{-1})^{-1/2} \geq \eta_\epsilon(\Lambda)$ for some $\epsilon \in (0, 1)$, then the statistical distance between $\mathcal{D}_{\Lambda, \sqrt{\Sigma_1}} + \mathcal{D}_{\Lambda, \sqrt{\Sigma_2}, \mathbf{c}}$ and $\mathcal{D}_{\Lambda, \sqrt{\Sigma}, \mathbf{c}}$ is at most $\frac{2\epsilon}{1-\epsilon}$, where $\Sigma = \Sigma_1 + \Sigma_2$.*

Proof. Since $\mathcal{D}_{\Lambda, \sqrt{\Sigma_1}} + \mathcal{D}_{\Lambda, \sqrt{\Sigma_2}, \mathbf{c}} \equiv \mathcal{D}_{\Lambda, \sqrt{\Sigma_1}} + \mathcal{D}_{\Lambda - \mathbf{c}, \sqrt{\Sigma_2}} + \mathbf{c}$ and similarly $\mathcal{D}_{\Lambda, \sqrt{\Sigma}, \mathbf{c}} \equiv \mathcal{D}_{\Lambda - \mathbf{c}, \sqrt{\Sigma}} + \mathbf{c}$, it suffices to bound the statistical distance between the distributions $\mathcal{D}_{\Lambda, \sqrt{\Sigma_1}} + \mathcal{D}_{\Lambda - \mathbf{c}, \sqrt{\Sigma_2}}$ and $\mathcal{D}_{\Lambda - \mathbf{c}, \sqrt{\Sigma}}$.

Write $\Sigma_i = \mathbf{S}_i \mathbf{S}_i^\top$ for $i = 1, 2$. We apply Lemma 5 with $m = 2$, $\Lambda_1 = \Lambda$, $\Lambda_2 = \Lambda - \mathbf{c}$, and $\mathbf{z} = (1, 1)^\top$, so that it suffices to show $\text{diag}(\mathbf{S}_1, \mathbf{S}_2) \geq \eta_\epsilon(\ker((1, 1)^\top \otimes \mathbf{I}_n) \cap (\Lambda \times \Lambda))$. Since $\mathbf{S}_i \geq s_{\min}(\mathbf{S}_i) \cdot \mathbf{I}$ for $i = 1, 2$, the condition is implied by $\text{diag}(s_{\min}(\mathbf{S}_1) \cdot \mathbf{I}, s_{\min}(\mathbf{S}_2) \cdot \mathbf{I}) \geq \eta_\epsilon(\ker((1, 1)^\top \otimes \mathbf{I}_n) \cap (\Lambda \times \Lambda))$. Then by Lemma 6, the latter is implied by $(s_{\min}(\mathbf{S}_1)^{-2} + s_{\min}(\mathbf{S}_2)^{-2})^{-1/2} = (s_{\min}(\Sigma_1)^{-1} + s_{\min}(\Sigma_2)^{-1})^{-1/2} \geq \eta_\epsilon(\Lambda)$. $\square$

Computational assumption. We state a generalisation of the LWE assumption, where the LWE matrix $\mathbf{A}$, secret $\mathbf{s}$ and error $\mathbf{e}$ are sampled from distributions $D_{\mathbf{A}}$, $D_{\mathbf{s}}$ and $D_{\mathbf{e}}$ respectively which are parameters of the assumption. The matrix $\mathbf{A}$ can be arbitrarily structured, which captures e.g. Power Ring-LWE [ARS24] and more generally (the natural LWE analogue of) Vanishing SIS [CLM23]. The adversary is furthermore given access to an oracle that can depend on $\mathbf{A}$, which captures hinted (not in the sense of Hint-MLWE) variants of LWE where non-trivial information about $\mathbf{A}$, e.g. short preimages of some target images with respect to $\mathbf{A}$, which may be available to the adversary. For example, similar flavour of hints are present in e.g. k -LWE [LPSS14], ℓ -Succinct LWE [Wee24], (the post-condition of) Evasive LWE [Wee22] and (the natural LWE analogues of) kRISIS [ACL⁺22] and (Strong) Hinted Vanishing SIS [DKLW25].

Definition 2 (LWE assumption). Let $\text{params} = (\mathcal{R}, n, m, q, D_{\mathbf{A}}, D_{\mathbf{s}}, D_{\mathbf{e}})$ be parametrised by λ , where $\mathcal{R}$ is a ring, $n, m, q \in \mathbb{N}$, $D_{\mathbf{A}}$ is a distribution over tuples $(\mathbf{A}, \mathbf{O})$, where $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ and $\mathbf{O}$ is an oracle⁶, $D_{\mathbf{s}}, D_{\mathbf{e}}$ are distributions over $\mathcal{R}^n$ and $\mathcal{R}^m$ respectively. For $b \in \{0, 1\}$ and any PPT $\mathcal{A}$, define the following security experiment:

Exp-LWE_{params, $\mathcal{A}$} ^b(1 ^{λ})

$(\mathbf{A}, \mathbf{O}) \leftarrow D_{\mathbf{A}} \sim \mathcal{R}_q^{n \times m} \times \{0, 1\}^*$
 $\mathbf{s} \leftarrow \$ D_{\mathbf{s}} \sim \mathcal{R}^n; \mathbf{e} \leftarrow \$ D_{\mathbf{e}} \sim \mathcal{R}^m$
if $b = 0$ **then** $\mathbf{b}^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \bmod q$
if $b = 1$ **then** $\mathbf{b} \leftarrow \$ \mathcal{R}_q^m$
return $\mathcal{A}^{\mathbf{O}}(\mathbf{A}, \mathbf{b})$

⁶ $\mathbf{O}$ potentially depends on $\mathbf{A}$.

The $\text{LWE}_{\text{params}}$ assumption states that for any PPT $\mathcal{A}$ it holds that

$$\begin{aligned} \text{Adv}_{\text{params}, \mathcal{A}}^{\text{LWE}}(\lambda) &:= |\Pr[\text{Exp-LWE}_{\text{params}, \mathcal{A}}^0(1^\lambda) = 1] - \Pr[\text{Exp-LWE}_{\text{params}, \mathcal{A}}^1(1^\lambda) = 1]| \\ &\leq \text{negl}(\lambda). \end{aligned}$$

When $D_{\mathbf{A}}$ and $D_{\mathbf{s}}$ are the uniform distribution over $\mathcal{R}_q^{n \times m}$ (with empty oracle $\mathcal{O}$) and $\mathcal{R}_q^n$ respectively, we omit them from params .

3 Core Statistical Claims

The goal of this section is to prove the core statistical claim given in Theorem 1 and its consequence given in Theorem 2, the latter of which will be used to prove the reduction from LWE to Leaky LWE in Theorem 3 in Section 4.

Theorem 1 below is a generalisation of [KLSS23, Lemma 7], which relaxes $\mathbf{L}$ from the structure $\mathbf{r}^T \otimes \mathbf{I}_{n+m}$ for a Gaussian vector $\mathbf{r}$ to an arbitrary matrix. Similar idea on linear transforms of Gaussians can be found in [DKL⁺23], with the difference that the latter argued the closeness of distributions via continuous Gaussians.

Theorem 1. *Let $m_0, m_1 \in \mathbb{N}$, $\Lambda_0 \subseteq \mathbb{R}^{m_0}$, $\Lambda_1 \subseteq \mathbb{R}^{m_1}$ be full-rank lattices, $\Sigma_0 \in \mathbb{R}^{m_0 \times m_0}$ and $\Sigma_1 \in \mathbb{R}^{m_1 \times m_1}$ be positive definite matrices and $\mathbf{L} \in \mathbb{Z}^{m_0 \times m_1}$ be any matrix satisfying $\mathbf{L}^T \Lambda_0 \subseteq \Lambda_1$. Define the following matrices (dependent on $\mathbf{L}, \Sigma_0, \Sigma_1$):*

$$\begin{aligned} \Gamma &:= \mathbf{L} \Sigma_1^{-1} && \in \mathbb{R}^{m_0 \times m_1}, \\ \Theta_0^{-1} &:= \Sigma_0^{-1} + \mathbf{L} \Sigma_1^{-1} \mathbf{L}^T && \in \mathbb{R}^{m_0 \times m_0}, \\ \Theta_1^{-1} &:= \Sigma_1^{-1} - \Gamma^T \Theta_0 \Gamma && \in \mathbb{R}^{m_1 \times m_1}. \end{aligned}$$

Then the distributions

$$\mathcal{D}_1 := \left\{ (\mathbf{x}_0, \mathbf{L}^T \mathbf{x}_0 + \mathbf{x}_1) \left| \begin{array}{l} \mathbf{x}_0 \leftarrow \mathcal{D}_{\Lambda_0, \sqrt{\Sigma_0}} \\ \mathbf{x}_1 \leftarrow \mathcal{D}_{\Lambda_1, \sqrt{\Sigma_1}} \end{array} \right. \right\} \quad \text{and} \quad \mathcal{D}_2 := \left\{ (\mathbf{y}_0, \mathbf{y}_1) \left| \begin{array}{l} \mathbf{y}_1 \leftarrow \mathcal{D}_{\Lambda_1, \sqrt{\Theta_1}} \\ \mathbf{c} := \Theta_0 \cdot \Gamma \cdot \mathbf{y}_1 \\ \mathbf{y}_0 \leftarrow \mathcal{D}_{\Lambda_0, \sqrt{\Theta_0, \mathbf{c}}} \end{array} \right. \right\}$$

are identical. Moreover, it holds that

$$\Theta_0 \geq (s_{\min}(\Sigma_0)^{-1} + s_{\max}(\mathbf{L} \mathbf{L}^T) \cdot s_{\min}(\Sigma_1)^{-1})^{-1}.$$

Proof. Write

$$(\mathbf{x}_0, \mathbf{L}^T \mathbf{x}_0 + \mathbf{x}_1) = \begin{pmatrix} \mathbf{I}_{m_0} & \\ \mathbf{L}^T & \mathbf{I}_{m_1} \end{pmatrix} \begin{pmatrix} \mathbf{x}_0 \\ \mathbf{x}_1 \end{pmatrix}.$$

Observe that $\begin{pmatrix} \mathbf{I}_{m_0} & \\ \mathbf{L}^T & \mathbf{I}_{m_1} \end{pmatrix}$ is invertible over $\mathbb{R}$ and hence defines an injective linear map via left-multiplication. Moreover, since $\mathbf{L}^T \Lambda_0 \subseteq \Lambda_1$, we have that

$$\begin{pmatrix} \mathbf{I}_{m_0} & \\ \mathbf{L}^T & \mathbf{I}_{m_1} \end{pmatrix} (\Lambda_0 \oplus \Lambda_1) = \Lambda_0 \oplus \Lambda_1.$$

Define the matrix

$$\Phi := \begin{pmatrix} \mathbf{I}_{m_0} & \\ \mathbf{L}^T & \mathbf{I}_{m_1} \end{pmatrix} \begin{pmatrix} \Sigma_0 & \\ & \Sigma_1 \end{pmatrix} \begin{pmatrix} \mathbf{I}_{m_0} & \mathbf{L} \\ & \mathbf{I}_{m_1} \end{pmatrix} = \begin{pmatrix} \Sigma_0 & \Sigma_0 \mathbf{L} \\ \mathbf{L}^T \Sigma_0 & \mathbf{L}^T \Sigma_0 \mathbf{L} + \Sigma_1 \end{pmatrix}.$$

Applying Lemma 4, we have that $\mathcal{D}_1$ and $\mathcal{D}_{\Lambda_0 \oplus \Lambda_1, \sqrt{\Phi}}$ are identical.

Next, we show that $\mathcal{D}_2$ is also identical to $\mathcal{D}_{\Lambda_0 \oplus \Lambda_1, \sqrt{\Phi}}$. Clearly, both distributions are supported by $\Lambda_0 \oplus \Lambda_1$. For each $(\mathbf{y}_0, \mathbf{y}_1) \in \Lambda_0 \oplus \Lambda_1$, we examine $\mathcal{D}_{\Lambda_0 \oplus \Lambda_1, \sqrt{\Phi}}(\mathbf{y}_0, \mathbf{y}_1)$. Noting that the inverse of Φ is given by

$$\Phi^{-1} = \begin{pmatrix} \Theta_0^{-1} & -\Gamma \\ -\Gamma^T & \Sigma_1^{-1} \end{pmatrix},$$

we have

$$\begin{aligned} \mathcal{D}_{\Lambda_0 \oplus \Lambda_1, \sqrt{\Phi}}(\mathbf{y}_0, \mathbf{y}_1) &\propto \exp(-\pi(\mathbf{y}_0^T, \mathbf{y}_1^T) \Phi^{-1} (\mathbf{y}_0, \mathbf{y}_1)) \\ &= \exp(-\pi(\mathbf{y}_0^T \Theta_0^{-1} \mathbf{y}_0 - \mathbf{y}_0^T \Gamma \mathbf{y}_1 - \mathbf{y}_1^T \Gamma^T \mathbf{y}_0 + \mathbf{y}_1^T \Sigma_1^{-1} \mathbf{y}_1)) \\ &= \exp(-\pi((\mathbf{y}_0 - \mathbf{c})^T \Theta_0^{-1} (\mathbf{y}_0 - \mathbf{c}) + \mathbf{y}_1^T (\Sigma_1^{-1} - \Gamma^T \Theta_0 \Gamma) \mathbf{y}_1)) \\ &= \exp(-\pi(\mathbf{y}_0 - \mathbf{c})^T \Theta_0^{-1} (\mathbf{y}_0 - \mathbf{c})) \cdot \exp(-\pi \mathbf{y}_1^T \Theta_1^{-1} \mathbf{y}_1) \\ &\propto \mathcal{D}_{\Lambda_0, \sqrt{\Theta_0}, \mathbf{c}}(\mathbf{y}_0) \cdot \mathcal{D}_{\Lambda_1, \sqrt{\Theta_1}}(\mathbf{y}_1) \\ &= \mathcal{D}_2(\mathbf{y}_0, \mathbf{y}_1) \end{aligned}$$

where $\mathbf{c} = \Theta_0 \cdot \Gamma \cdot \mathbf{y}_1$.

Finally, to lower bound Θ_0 , we instead upper bound $\Theta_0^{-1} = \Sigma_0^{-1} + \mathbf{L} \Sigma_1^{-1} \mathbf{L}^T$ by

$$\begin{aligned} \Theta_0^{-1} &\leq s_{\max}(\Sigma_0^{-1}) + s_{\max}(\mathbf{L} \Sigma_1^{-1} \mathbf{L}^T) \\ &\leq s_{\max}(\Sigma_0^{-1}) + s_{\max}(\mathbf{L} \mathbf{L}^T) \cdot s_{\max}(\Sigma_1^{-1}) \\ &= s_{\min}(\Sigma_0)^{-1} + s_{\max}(\mathbf{L} \mathbf{L}^T) \cdot s_{\min}(\Sigma_1)^{-1}. \end{aligned}$$

The lower bound of Θ_0 follows. $\square$

Theorem 2 below is analogous to [DKL⁺23, Theorem 2]. The difference with Theorem 1 is that the vector $\mathbf{y}_0$ is replaced by a sum of two vectors $\mathbf{y}_{00}, \mathbf{y}_{01}$. This will be useful for our subsequent reduction for Leaky LWE, where $\mathbf{y}_{00}$ will be the LWE error of the underlying LWE problem, and $\mathbf{y}_{01}$ will be sampled by the reduction and added to (the LWE sample with) the error $\mathbf{y}_{00}$.

Theorem 2. *Let $m_0, m_1 \in \mathbb{N}$ and $\Lambda_0 \subseteq \mathbb{R}^{m_0}$, $\Lambda_1 \subseteq \mathbb{R}^{m_1}$ be full-rank lattices. Let $\Theta_{00}, \Sigma_0 \in \mathbb{R}^{m_0 \times m_0}$ and $\Sigma_1 \in \mathbb{R}^{m_1 \times m_1}$ be positive definite matrices. Fix any $\mathbf{L} \in \mathbb{Z}^{m_0 \times m_1}$ satisfying $\mathbf{L} \Lambda_0 \subseteq \Lambda_1$, and define the following matrices (dependent on $\mathbf{L}, \Theta_{00}, \Sigma_0, \Sigma_1$):*

$$\begin{aligned} \Gamma &:= \mathbf{L} \Sigma_1^{-1} && \in \mathbb{R}^{m_0 \times m_1}, \\ \Theta_0^{-1} &:= \Sigma_0^{-1} + \mathbf{L} \Sigma_1^{-1} \mathbf{L}^T && \in \mathbb{R}^{m_0 \times m_0}, \\ \Theta_1^{-1} &:= \Sigma_1^{-1} - \Gamma^T \Theta_0 \Gamma && \in \mathbb{R}^{m_1 \times m_1}, \\ \Theta_{01} &:= \Theta_0 - \Theta_{00} && \in \mathbb{R}^{m_0 \times m_0}. \end{aligned}$$

Let $\epsilon \in (0, 1)$ and $\gamma^{-1} := s_{\min}(\Sigma_0)^{-1} + s_{\max}(\mathbf{L} \mathbf{L}^T) \cdot s_{\min}(\Sigma_1)^{-1}$. Suppose

- $\Theta_{00} < \gamma$, and
- $(s_{\min}(\Theta_{00})^{-1} + (\gamma - s_{\max}(\Theta_{00}))^{-1})^{-1/2} \geq \eta_\epsilon(\Lambda_0)$.

Then the statistical distance between the distributions

$$\left\{ (\mathbf{x}_0, \mathbf{L}^T \mathbf{x}_0 + \mathbf{x}_1) \left| \begin{array}{l} \mathbf{x}_0 \leftarrow \mathcal{D}_{\Lambda_0, \sqrt{\Sigma_0}} \\ \mathbf{x}_1 \leftarrow \mathcal{D}_{\Lambda_1, \sqrt{\Sigma_1}} \end{array} \right. \right\} \quad \text{and} \quad \left\{ (\mathbf{y}_{00} + \mathbf{y}_{01}, \mathbf{y}_1) \left| \begin{array}{l} \mathbf{y}_1 \leftarrow \mathcal{D}_{\Lambda_1, \sqrt{\Theta_1}} \\ \mathbf{c} := \Theta_0 \cdot \Gamma \cdot \mathbf{y}_1 \\ \mathbf{y}_{00} \leftarrow \mathcal{D}_{\Lambda_0, \sqrt{\Theta_{00}}} \\ \mathbf{y}_{01} \leftarrow \mathcal{D}_{\Lambda_0, \sqrt{\Theta_{01}, \mathbf{c}}} \end{array} \right. \right\}$$

is at most $\frac{2\epsilon}{1-\epsilon}$.

Proof. It suffices to show that, for any fixed $\mathbf{c} \in \mathbb{R}^{m_0}$, the statistical distance between

$$\{\mathbf{y}_0 \mid \mathbf{y}_0 \leftarrow \$ \mathcal{D}_{\Lambda_0, \sqrt{\mathbf{\Theta}_0}, \mathbf{c}}\} \quad \text{and} \quad \{\mathbf{y}_{00} + \mathbf{y}_{01} \mid \mathbf{y}_{00} \leftarrow \$ \mathcal{D}_{\Lambda_0, \sqrt{\mathbf{\Theta}_{00}}}, \mathbf{y}_{01} \leftarrow \$ \mathcal{D}_{\Lambda_0, \sqrt{\mathbf{\Theta}_{01}}, \mathbf{c}}\},$$

or in short, that between $\mathcal{D}_{\Lambda_0, \sqrt{\mathbf{\Theta}_0}, \mathbf{c}}$ and $\mathcal{D}_{\Lambda_0, \sqrt{\mathbf{\Theta}_{00}}} + \mathcal{D}_{\Lambda_0, \sqrt{\mathbf{\Theta}_{01}}, \mathbf{c}}$, is bounded by $\frac{2\epsilon}{1-\epsilon}$, then the claim follows from invoking Theorem 1.

To show the above, we simply apply Lemma 7, for which we check that all conditions required therein are satisfied. It holds that $\mathbf{\Theta}_{00} > 0$ and $\mathbf{\Theta}_{00} + \mathbf{\Theta}_{01} = \mathbf{\Theta}_0$ by design, so it remains to show that (1) $\mathbf{\Theta}_{01} > 0$ and (2) $(s_{\min}(\mathbf{\Theta}_{00})^{-1} + s_{\min}(\mathbf{\Theta}_{01})^{-1})^{-1/2} \geq \eta_\epsilon(\Lambda_0)$. Recall that by Theorem 1 we have the guarantee of

$$\mathbf{\Theta}_0 \geq (s_{\min}(\mathbf{\Sigma}_0)^{-1} + s_{\max}(\mathbf{L}\mathbf{L}^T) \cdot s_{\min}(\mathbf{\Sigma}_1)^{-1})^{-1} = \gamma. \quad (2)$$

Therefore

$$\mathbf{\Theta}_{01} = \mathbf{\Theta}_0 - \mathbf{\Theta}_{00} \geq \gamma \cdot \mathbf{I} - \mathbf{\Theta}_{00} > 0,$$

where the first inequality also implies

$$s_{\min}(\mathbf{\Theta}_{01}) \geq \gamma - s_{\max}(\mathbf{\Theta}_{00}).$$

Now since $(\sigma_1^{-1} + \sigma_2^{-1})^{-1/2}$ is an increasing function in σ_2 for any fixed $\sigma_1 > 0$, we have

$$(s_{\min}(\mathbf{\Theta}_{00})^{-1} + s_{\min}(\mathbf{\Theta}_{01})^{-1})^{-1/2} \geq (s_{\min}(\mathbf{\Theta}_{00})^{-1} + (\gamma - s_{\max}(\mathbf{\Theta}_{00}))^{-1})^{-1/2} \geq \eta_\epsilon(\Lambda_0),$$

as required. $\square$

Remark 1. In Theorem 2, a sufficient condition for the two constraints on $\mathbf{\Theta}_{00}$ and γ to hold is to have $\min\{s_{\min}(\mathbf{\Theta}_{00}), \gamma - s_{\max}(\mathbf{\Theta}_{00})\} \geq 2 \cdot \eta_\epsilon(\Lambda_0)^2$.

Theorem 2 simplifies [DKL⁺23, Theorem 2] by avoiding additional continuous Gaussian distributions. As a consequence of directly handling discrete Gaussians, we also entirely remove the variables s, t (and hence their resulting multiplicative factors) and constants $1/4, 3/4, 2$, etc. in [DKL⁺23, Theorem 2]. Our Theorem 2 captures a slightly wider range of parameters, due to tighter bounds on singular values in the proof. In particular, [DKL⁺23, Theorem 2] requires to lower-bound $s_{\min}(\mathbf{\Sigma}_1)$ by $\approx s_{\max}(\mathbf{L}\mathbf{L}^T) \cdot s_{\max}(\mathbf{\Sigma}_0)$, which results in a larger $\mathbf{\Sigma}_1$ when $s_{\max}(\mathbf{\Sigma}_0)$ is larger. In contrast, in Theorem 2 above, by expanding the condition $\mathbf{\Theta}_{00} < \gamma$, one obtains a lower bound of $s_{\min}(\mathbf{\Sigma}_1) > \frac{s_{\max}(\mathbf{L}\mathbf{L}^T)}{\mathbf{\Theta}_{00}^{-1} - s_{\min}(\mathbf{\Sigma}_0)^{-1}}$. When keeping $s_{\max}(\mathbf{L}\mathbf{L}^T), \mathbf{\Theta}_{00}$ unchanged, the bound implies that it is possible to increase $s_{\min}(\mathbf{\Sigma}_0)$ in exchange for a less stringent lower bound for $s_{\min}(\mathbf{\Sigma}_1)$, a trade-off not achieved by [DKL⁺23, Theorem 2]. Moreover, the said lower bound only depends on $s_{\min}(\mathbf{\Sigma}_0)$ but not $s_{\max}(\mathbf{\Sigma}_0)$, hence Theorem 2 may support a smaller $\mathbf{\Sigma}_1$ when $\mathbf{\Sigma}_0$ is skewed (up to other appropriate parameters), i.e. the difference of $s_{\max}(\mathbf{\Sigma}_0)$ and $s_{\min}(\mathbf{\Sigma}_0)$ is large. We discuss a gap of the proof of [DKL⁺23, Theorem 2] in Section A.

4 Leaky LWE and Applications

We introduce the Leaky LWE assumption, and show how the statistical lemmas in Section 3 can be used to prove the hardness of Leaky LWE from standard LWE.

Definition 3 (Leaky LWE assumption). Let $\text{params} = (\mathcal{R}, n, m, q, Q, D_{\mathbf{A}}, D_{\mathbf{s}}, D_{\mathbf{e}}, D_{\mathbf{f}}, \mathcal{L})$ be parametrised by λ , where $\mathcal{R}$ is a ring, $n, m, q, Q \in \mathbb{N}$, $D_{\mathbf{A}}$ is a distribution over tuples $(\mathbf{A}, \mathbf{O})$, where $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ and $\mathbf{O} = \mathbf{O}_{\mathbf{A}}$ is an oracle, $D_{\mathbf{s}}, D_{\mathbf{e}}, D_{\mathbf{f}}$ are distributions over $\mathcal{R}^n, \mathcal{R}^m$ and $\mathbb{Z}^Q$ respectively, and $\mathcal{L} \subseteq \mathcal{R}^{n \times Q} \times \mathcal{R}^{m \times Q}$ is a polynomial-time decidable family of tuple of matrices. For $b \in \{0, 1\}$ and any adversary $\mathcal{A} = (\mathcal{A}_0, \mathcal{A}_1)$, define the following security experiment:

```

Exp-LLWEparams, Ab(1λ)
(A, O) ← DA ∼ Rqn×m × {0, 1}*
(Ls, Le, state) ← A0O(A); assert (Ls, Le) ∈ L
s ←$ Ds ∼ Rn; e ←$ De ∼ Rm; f ←$ Df ∼ RQ
if b = 0 then bT := sTA + eT mod q
if b = 1 then b ←$ Rqm
lT := (sT, eT)  $\begin{pmatrix} \mathbf{L}_s \\ \mathbf{L}_e \end{pmatrix}$  + fT
return A1O(b, l, state)

```

The LLWE_{params} assumption states that for any PPT $\mathcal{A}$ it holds that

$$\text{Adv}_{\text{params}, \mathcal{A}}^{\text{LLWE}}(\lambda) := |\Pr[\text{Exp-LLWE}_{\text{params}, \mathcal{A}}^0(1^\lambda) = 1] - \Pr[\text{Exp-LLWE}_{\text{params}, \mathcal{A}}^1(1^\lambda) = 1]| \leq \text{negl}(\lambda).$$

When D_A and D_s are the uniform distribution over $\mathcal{R}_q^{n \times m}$ (with an empty oracle O) and $\mathcal{R}_q^n$ respectively, we omit them from params .

By restricting Definition 3 to support only $\mathcal{L} \subseteq \emptyset \times \mathcal{R}_q^{m \times Q}$, i.e. there is no leakage on the LWE secret $\mathbf{s}$, we recover the error-leakage LWE assumption of [DKL⁺23]. Similarly, by restricting $\mathcal{L}$ to contain tuple of matrices $(\mathbf{L}_s, \mathbf{L}_e)$ of the form

$$\begin{pmatrix} \mathbf{L}_s \\ \mathbf{L}_e \end{pmatrix} = \mathbf{r}^T \otimes \mathbf{I}_{n+m}$$

where $\mathbf{r} \in \mathcal{R}_q^Q$ are norm-bounded elements, we recover (an assumption which implies⁷) the Hint-MLWE assumption [KLSS23].

4.1 Reduction from LWE

We show that the Leaky LWE assumption, with a possibly structured matrices $\mathbf{A}$ and correlated oracles O , is implied by the LWE assumption with the same distribution of structured matrices and oracles. In particular, the “standard” Leaky LWE assumption with uniform matrices $\mathbf{A}$ and without any oracles is implied by the standard LWE assumption. Throughout the rest, we will consider two parameters sets, given by Conditions 1 and 2 below. The superscript $*$ is used to denote parameters of the underlying LWE problem.

- Condition 1 concerns LWE and Leaky LWE parameters where the LWE secrets follow discrete Gaussians. This setting allows Leaky LWE with leakages on both the LWE secret $\mathbf{s}$ and error $\mathbf{e}$ (see definition of $\mathcal{L}$).
- Condition 2 concerns LWE and Leaky LWE parameters where the LWE secret distributions are identical and arbitrary, but restricts Leaky LWE to support leakages only on the LWE error $\mathbf{e}$ (see definition of $\mathcal{L}$).

Condition 1. *The parameters*

$$\text{params}^* := (\mathcal{R}, n, m, q, D_A, D_s^*, D_e^*), \quad \text{params} := (\mathcal{R}, n, m, q, Q, D_A, D_s, D_e, D_f, \mathcal{L})$$

and ϵ are parametrised by λ , where

- $D_s^* = \mathcal{D}_{\mathcal{R}^n, \sqrt{\Sigma_s^*}}$ and $D_s = \mathcal{D}_{\mathcal{R}^n, \sqrt{\Sigma_s}}$ where $\Sigma_s^*, \Sigma_s \in \mathbb{R}^{\varphi n \times \varphi n}$,

⁷In Hint-MLWE, the leakage matrices $\mathbf{L}_s, \mathbf{L}_e$ are sampled by the experiment.

- $D_{\mathbf{e}}^* = \mathcal{D}_{\mathcal{R}^m, \sqrt{\Sigma_{\mathbf{e}}^*}}$ and $D_{\mathbf{e}} = \mathcal{D}_{\mathcal{R}^m, \sqrt{\Sigma_{\mathbf{e}}}}$ where $\Sigma_{\mathbf{e}}^*, \Sigma_{\mathbf{e}} \in \mathbb{R}^{\varphi m \times \varphi m}$,
- $D_{\mathbf{f}} = \mathcal{D}_{\mathcal{R}^m, \sqrt{\Sigma_{\mathbf{f}}}}$ where $\Sigma_{\mathbf{f}} \in \mathbb{R}^{\varphi Q \times \varphi Q}$,
- $s_{\min}(\text{diag}(\Sigma_{\mathbf{s}}^*, \Sigma_{\mathbf{e}}^*)) \geq 2\eta_{\epsilon}(\mathcal{R}^{n+m})^2$,
- $\mathcal{L} := \{\mathbf{L} = (\mathbf{L}_{\mathbf{s}}, \mathbf{L}_{\mathbf{e}}) \in \mathcal{R}^{n \times Q} \times \mathcal{R}^{m \times Q} : \|\mathbf{L}^T \mathbf{L}\| \leq \beta\}$, where

$$\beta = s_{\min}(\Sigma_{\mathbf{f}}) \cdot ((2\eta_{\epsilon}(\mathcal{R}^{n+m})^2 + s_{\max}(\text{diag}(\Sigma_{\mathbf{s}}^*, \Sigma_{\mathbf{e}}^*)))^{-1} - s_{\min}(\text{diag}(\Sigma_{\mathbf{s}}, \Sigma_{\mathbf{e}}))^{-1}).$$

Example 1. The following set of parameters satisfy Condition 1: $D_{\mathbf{s}}^* = \mathcal{D}_{\mathcal{R}^n, \sigma^*}$, $D_{\mathbf{s}} = \mathcal{D}_{\mathcal{R}^n, \sigma}$, $D_{\mathbf{e}}^* = \mathcal{D}_{\mathcal{R}^m, \sigma^*}$, $D_{\mathbf{e}} = \mathcal{D}_{\mathcal{R}^m, \sigma}$, $D_{\mathbf{f}} = \mathcal{D}_{\mathcal{R}^m, \sigma_{\mathbf{f}}}$, $\sigma^* \geq \sqrt{2}\eta_{\epsilon}(\mathcal{R}^{n+m})$,

$$\mathcal{L} := \{\mathbf{L} = (\mathbf{L}_{\mathbf{s}}, \mathbf{L}_{\mathbf{e}}) \in \mathcal{R}^{n \times Q} \times \mathcal{R}^{m \times Q} : \|\mathbf{L}^T \mathbf{L}\| \leq \beta\},$$

where

$$\beta = \sigma_{\mathbf{f}}^2 \cdot ((2\eta_{\epsilon}(\mathcal{R}^{n+m})^2 + (\sigma^*)^2)^{-1} - \sigma^{-2}).$$

Condition 2. The parameters

$$\text{params}^* := (\mathcal{R}, n, m, q, D_{\mathbf{A}}, D_{\mathbf{s}}^*, D_{\mathbf{e}}^*), \quad \text{params} := (\mathcal{R}, n, m, q, Q, D_{\mathbf{A}}, D_{\mathbf{s}}, D_{\mathbf{e}}, D_{\mathbf{f}}, \mathcal{L})$$

and ϵ are parametrised by λ , where

- $D_{\mathbf{s}}^* \equiv D_{\mathbf{s}}$ is any distribution over $\mathcal{R}^n$,
- $D_{\mathbf{e}}^* = \mathcal{D}_{\mathcal{R}^m, \sqrt{\Sigma_{\mathbf{e}}^*}}$ and $D_{\mathbf{e}} = \mathcal{D}_{\mathcal{R}^m, \sqrt{\Sigma_{\mathbf{e}}}}$ where $\Sigma_{\mathbf{e}}^*, \Sigma_{\mathbf{e}} \in \mathbb{R}^{\varphi m \times \varphi m}$,
- $D_{\mathbf{f}} = \mathcal{D}_{\mathcal{R}^m, \sqrt{\Sigma_{\mathbf{f}}}}$ where $\Sigma_{\mathbf{f}} \in \mathbb{R}^{\varphi Q \times \varphi Q}$,
- $s_{\min}(\Sigma_{\mathbf{e}}^*) \geq 2\eta_{\epsilon}(\mathcal{R}^m)^2$,
- $\mathcal{L} := \{\mathbf{L} = (*, \mathbf{L}_{\mathbf{e}}) \in \emptyset \times \mathcal{R}^{m \times Q} : \|\mathbf{L}^T \mathbf{L}\| \leq \beta\}$, where

$$\beta = s_{\min}(\Sigma_{\mathbf{f}}) \cdot ((2\eta_{\epsilon}(\mathcal{R}^m)^2 + s_{\max}(\Sigma_{\mathbf{e}}^*))^{-1} - s_{\min}(\Sigma_{\mathbf{e}})^{-1}).$$

Example 2. The following set of parameters satisfy Condition 2: $D_{\mathbf{s}}^* = D_{\mathbf{s}} = \mathcal{U}(\mathcal{R}^n)$, $D_{\mathbf{e}}^* = \mathcal{D}_{\mathcal{R}^m, \sigma^*}$, $D_{\mathbf{e}} = \mathcal{D}_{\mathcal{R}^m, \sigma}$, $D_{\mathbf{f}} = \mathcal{D}_{\mathcal{R}^m, \sigma_{\mathbf{f}}}$, $\sigma^* \geq \sqrt{2}\eta_{\epsilon}(\mathcal{R}^{n+m})$,

$$\mathcal{L} := \{\mathbf{L} = (*, \mathbf{L}_{\mathbf{e}}) \in \emptyset \times \mathcal{R}^{m \times Q} : \|\mathbf{L}^T \mathbf{L}\| \leq \beta\},$$

where

$$\beta = \sigma_{\mathbf{f}}^2 \cdot ((2\eta_{\epsilon}(\mathcal{R}^m)^2 + (\sigma^*)^2)^{-1} - \sigma^{-2}).$$

Both Conditions 1 and 2 above are stated with full generality, but for typical applications we believe the special cases in Examples 1 and 2 suffice. In both, the major constraint is an upper bound on $\|\mathbf{L}^T \mathbf{L}\|$ which is roughly $\sigma_{\mathbf{f}}^2$, where $\sigma_{\mathbf{f}}$ is the Gaussian parameter of the vector $\mathbf{f}$. Formally, the constraints come from Theorem 1, which are needed for arguing closeness of linear transforms of Gaussians. Intuitively, since the leakage in Leaky LWE is $\mathbf{e}^T \mathbf{L} + \mathbf{f}^T$, $\|\mathbf{f}\|$ can not be too small relatively to $\|\mathbf{L}\|$, as otherwise an adversary may learn too much information about $\mathbf{e}^T \mathbf{L}$, from which it may potentially recover $\mathbf{e}$.⁸

⁸[AP12] showed that if $\sigma_{\mathbf{f}}$ is constant in the number of columns Q of $\mathbf{L}$, then there exists a large enough Q such that the assumption cannot hold. Their attack relies on upper-bounding the maximum singular value of the (right) pseudo-inverse of $\mathbf{L}$, equivalently lower-bounding $\|\mathbf{L}\|$. This means a lower-bound of $\sigma_{\mathbf{f}}$ dependent on $\|\mathbf{L}\|$ is necessary.

Theorem 3 (LWE $\implies$ Leaky LWE). *Let params^* , params and ϵ be such that all constraints in either Condition 1 or Condition 2 are satisfied. Then, for any PPT adversary $\mathcal{A}$, there exists a PPT adversary $\mathcal{B}$ such that*

$$\text{Adv}_{\text{params}^*, \mathcal{B}}^{\text{LWE}}(\lambda) \geq \text{Adv}_{\text{params}, \mathcal{A}}^{\text{LLWE}}(\lambda) - 4\epsilon/(1 - \epsilon).$$

Proof. We prove the claim with parameters under Condition 1, and sketch the difference with Condition 2 at the end.

To connect with the notation in Theorem 2, define the shorthands $\Theta_{00} := \text{diag}(\Sigma_s^*, \Sigma_e^*)$ and $\Sigma_0 := \text{diag}(\Sigma_s, \Sigma_e)$. Also let $\Sigma_1 := \Sigma_f$. Let $\mathcal{A}$ be a PPT adversary against $\text{LLWE}_{\text{params}}$, we construct the claimed PPT adversary $\mathcal{B}$. On input a $\text{LWE}_{\text{params}^*}$ challenge instance $(\mathbf{A}, \hat{\mathbf{b}}) \in \mathcal{R}_q^{n \times m} \times \mathcal{R}_q^m$, where $\hat{\mathbf{b}}^T$ is either $\hat{\mathbf{s}}^T \mathbf{A} + \hat{\mathbf{e}}^T \bmod q$ with $\begin{pmatrix} \hat{\mathbf{s}} \\ \hat{\mathbf{e}} \end{pmatrix} \leftarrow \mathcal{D}_{\mathcal{R}^{n+m}, \sqrt{\Theta_{00}}}$ or uniform over $\mathcal{R}_q^m$, let $\mathcal{B}$ proceed as follows:

- Pass $\mathbf{A}$ to $\mathcal{A}$, and receive from $\mathcal{A}$ its chosen leakage matrix $\mathbf{L}$. Assert $\mathbf{L} \in \mathcal{L}$. Throughout the experiment, for any query on oracle $\mathbf{O}$ made by $\mathcal{A}$, forward the query to the same $\mathbf{O}$ available to $\mathcal{B}$ from the $\text{LWE}_{\text{params}^*}$ experiment.
- Set the following parameters:

$$\begin{aligned} \Gamma &:= \Psi(\mathbf{L})\Sigma_1^{-1} && \in \mathbb{R}^{\varphi(n+m) \times \varphi Q}, \\ \Theta_0^{-1} &:= \Sigma_0^{-1} + \Psi(\mathbf{L})\Sigma_1^{-1}\Psi(\mathbf{L})^T && \in \mathbb{R}^{\varphi(n+m) \times \varphi(n+m)}, \\ \Theta_1^{-1} &:= \Sigma_1^{-1} - \Gamma^T \Theta_0 \Gamma && \in \mathbb{R}^{\varphi Q \times \varphi Q}, \\ \Theta_{01} &:= \Theta_0 - \Theta_{00} && \in \mathbb{R}^{\varphi(n+m) \times \varphi(n+m)}. \end{aligned}$$

(We recall that Ψ is the unique linear map induced by the embedding ψ , for details see Section 2.)

- Sample $\mathbf{l} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sqrt{\Theta_1}}$. Set $\mathbf{c} := \psi^{-1}(\Theta_0 \cdot \Gamma \cdot \psi(\mathbf{l}))$.⁹ Sample $\begin{pmatrix} \check{\mathbf{s}} \\ \check{\mathbf{e}} \end{pmatrix} \leftarrow \mathcal{D}_{\mathcal{R}^{n+m}, \sqrt{\Theta_{01} \cdot \mathbf{c}}}$.
- Compute $\mathbf{b}^T := \hat{\mathbf{b}}^T + \check{\mathbf{s}}^T \cdot \mathbf{A} + \check{\mathbf{e}}^T \bmod q$.
- Pass $(\mathbf{b}, \mathbf{l})$ to $\mathcal{B}$. Eventually return whatever $\mathcal{B}$ returns.

We analyse the output $(\mathbf{b}, \mathbf{l})$ to $\mathcal{B}$. If $\hat{\mathbf{b}}^T = \hat{\mathbf{s}}^T \mathbf{A} + \hat{\mathbf{e}}^T \bmod q$ where $\begin{pmatrix} \hat{\mathbf{s}} \\ \hat{\mathbf{e}} \end{pmatrix} \leftarrow \mathcal{D}_{\mathcal{R}^{n+m}, \sqrt{\Theta_{00}}}$, then we have

$$\begin{aligned} (\mathbf{b}^T, \mathbf{l}^T) &= (\hat{\mathbf{s}}^T \mathbf{A} + \hat{\mathbf{e}}^T + \check{\mathbf{s}}^T \mathbf{A} + \check{\mathbf{e}}^T \bmod q, \mathbf{l}^T) \\ &= ((\hat{\mathbf{s}}^T + \check{\mathbf{s}}^T) \cdot \mathbf{A} + (\hat{\mathbf{e}}^T + \check{\mathbf{e}}^T) \bmod q, \mathbf{l}^T) \\ &\approx_s (\mathbf{s}^T \mathbf{A} + \mathbf{e}^T \bmod q, (\mathbf{s}, \mathbf{e})^T \mathbf{L} + \mathbf{f}^T) \end{aligned}$$

for $\begin{pmatrix} \mathbf{s} \\ \mathbf{e} \end{pmatrix} \leftarrow \mathcal{D}_{\mathcal{R}^{n+m}, \sqrt{\Sigma_0}}$ and $\mathbf{f} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sqrt{\Sigma_1}}$, where the last line follows from Theorem 2 with statistical distance at most $\frac{2\epsilon}{1-\epsilon}$. Therefore, $\mathcal{B}$ statistically simulates the view of $\text{Exp-LLWE}_{\text{params}}^0$ to $\mathcal{A}$.

Else, if $\hat{\mathbf{b}}$ is uniform over $\mathcal{R}_q^m$, then

$$(\mathbf{b}^T, \mathbf{l}^T) = (\hat{\mathbf{b}}^T + \check{\mathbf{s}}^T \mathbf{A} + \check{\mathbf{e}}^T \bmod q, \mathbf{l}^T)$$

⁹Here, we are abusing notation slightly because $\Theta_0 \cdot \Gamma \cdot \psi(\mathbf{l})$ may not be in the image of ψ . However, this will not be a problem because we will only operate on $\psi(\mathbf{c})$.

$$\begin{aligned} &\equiv (\mathcal{U}(\mathcal{R}_q^n), \mathbf{1}^T) \\ &\approx_s (\mathcal{U}(\mathcal{R}_q^n), (\mathbf{s}, \mathbf{e})^T \mathbf{L} + \mathbf{f}^T) \end{aligned}$$

for $\begin{pmatrix} \mathbf{s} \\ \mathbf{e} \end{pmatrix} \leftarrow \mathcal{D}_{\mathcal{R}^{n+m}, \sqrt{\Sigma_0}}$ and $\mathbf{f} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sqrt{\Sigma_1}}$, where the last line again follows from Theorem 2 with the same reasoning as in the previous case with statistical distance at most $\frac{2\epsilon}{1-\epsilon}$. Therefore, $\mathcal{B}$ statistically simulates the view of $\text{Exp-LLWE}_{\text{params}}^1$ to $\mathcal{A}$.

The above concludes the proof with parameters under Condition 1. The proof with Condition 2 is a special case and is analogous with differences highlighted below. First, we redefine the shorthands $\Theta_{00} := \Sigma_{\mathbf{e}}^*$ and $\Sigma_0 := \Sigma_{\mathbf{e}}$. Upon receiving the leakage matrix $\mathbf{L} = (*, \mathbf{L}_{\mathbf{e}})$ from the adversary, in the second step of the reduction it sets

$$\begin{aligned} \Gamma &:= \Psi(\mathbf{L}_{\mathbf{e}}) \Sigma_1^{-1} && \in \mathbb{R}^{\varphi m \times \varphi Q}, \\ \Theta_0^{-1} &:= \Sigma_0^{-1} + \Psi(\mathbf{L}_{\mathbf{e}}) \Sigma_1^{-1} \Psi(\mathbf{L}_{\mathbf{e}})^T && \in \mathbb{R}^{\varphi m \times \varphi m}, \\ \Theta_1^{-1} &:= \Sigma_1^{-1} - \Gamma^T \Theta_0 \Gamma && \in \mathbb{R}^{\varphi Q \times \varphi Q}, \\ \Theta_{01} &:= \Theta_0 - \Theta_{00} && \in \mathbb{R}^{\varphi m \times \varphi m}. \end{aligned}$$

In the third step it samples $\check{\mathbf{e}} \leftarrow \mathcal{D}_{\mathcal{R}^m, \sqrt{\Theta_{01}}, \mathbf{c}}$ (where $\mathbf{c}$ defined by the new Θ_0 and Γ), and sets in the fourth step

$$\mathbf{b}^T := \hat{\mathbf{b}}^T + \check{\mathbf{e}}^T \bmod q.$$

If $\hat{\mathbf{b}}^T = \hat{\mathbf{s}}^T \mathbf{A} + \hat{\mathbf{e}}^T \bmod q$ where $\hat{\mathbf{e}} \leftarrow \mathcal{D}_{\mathcal{R}^m, \sqrt{\Theta_{00}}}$, then

$$(\mathbf{b}^T, \mathbf{1}^T) = (\hat{\mathbf{s}}^T \mathbf{A} + \hat{\mathbf{e}}^T + \check{\mathbf{e}}^T \bmod q, \mathbf{1}^T) \approx_s (\hat{\mathbf{s}}^T \mathbf{A} + \mathbf{e}^T \bmod q, \mathbf{e}^T \mathbf{L} + \mathbf{f}^T)$$

for $\mathbf{e} \leftarrow \mathcal{D}_{\mathcal{R}^m, \sqrt{\Sigma_0}}$ and $\mathbf{f} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sqrt{\Sigma_1}}$, where the last line follows from Theorem 2. Else, if $\hat{\mathbf{b}}$ is uniform, then

$$(\mathbf{b}^T, \mathbf{1}^T) = (\hat{\mathbf{b}}^T + \check{\mathbf{e}}^T \bmod q, \mathbf{1}^T) \equiv (\mathcal{U}(\mathcal{R}_q^n), \mathbf{1}^T) \approx_s (\mathcal{U}(\mathcal{R}_q^n), \mathbf{e}^T \mathbf{L} + \mathbf{f}^T),$$

for $\mathbf{e} \leftarrow \mathcal{D}_{\mathcal{R}^m, \sqrt{\Sigma_0}}$ and $\mathbf{f} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sqrt{\Sigma_1}}$ as desired. $\square$

Remark 2. Both Definition 3 and Theorem 3 can easily be generalised to support leakages with codomain $\mathbb{R}^{Q'}$ instead of $\mathbb{R}^Q$, i.e. leakages of the form $\mathbf{l} := \mathbf{L} \cdot \psi \begin{pmatrix} \mathbf{s} \\ \mathbf{e} \end{pmatrix} + \mathbf{f} \in \mathbb{R}^{Q'}$, where $\mathbf{L} \in \mathbb{R}^{Q' \times \varphi(n+m)}$ and $\mathbf{f} \in \mathbb{R}^{Q'}$. This is more expressive since now $\mathbf{L}$ is not necessarily in the image of Ψ .

Our reduction for Leaky LWE follows the same template as those for Error-Leakage LWE [DKL⁺23] and Hint-MLWE [KLSS23], and can be seen as a unification of the two.

Relative to Hint-MLWE ([KLSS23, Theorem 1]), Theorem 3 supports much more general parameters, such as $\mathcal{L}$ a set of arbitrary matrices of bounded norm, non-spherical Gaussian parameters, and arbitrary number ring equipped with an embedding. We envision these generalisations to enable broader applications, such as new lattice-based Σ -protocols not necessarily resembling the construction template of [KLSS23]. Similar to Hint-MLWE, leakages on the LWE secret $\mathbf{s}$ is supported only when $\mathbf{s}$ follows discrete Gaussians (Condition 1). When specialising to the particular choice of $\mathcal{L}$ in [KLSS23] and spherical Gaussians, the parameters supported by the reduction of Hint-MLWE [KLSS23, Theorem 1]¹⁰ are essentially a special case in the regime captured by Theorem 3.

Comparing to Error-Leakage LWE ([DKL⁺23, Theorem 3]), Theorem 3 additionally takes care of leakages on the LWE secret $\mathbf{s}$, while keeping the semi-adaptivity of the

¹⁰[KLSS23, Theorem 1] imposed the condition (adapted to our notation) $(\sigma^*)^{-2} = 2(\sigma_0^{-2} + B\sigma_1^{-2})$, where B corresponds to a norm bound on (some function of) $\mathbf{L}$. This constraint allowed to simplify some of their proof arguments, although it may be relaxed.

adversary on choosing $\mathbf{L}$. Beyond this, our improvements in Theorem 2 over [DKL⁺23, Theorem 2] immediately allow us to achieve the correspondingly improved reduction parameters, e.g. fewer constraints and overheads due to dropping continuous Gaussians, and more flexible choices of Σ_0, Σ_1 (see Section 3 for details). In Section A we discuss a few discrepancies in [DKL⁺23, Theorem 3] when invoking [DKL⁺23, Theorem 2].

4.2 How not to use statistical noise flooding

The indistinguishability of the pairs of distributions considered in Lemmas 8 and 9 are typically argued via noise flooding with super-polynomial modulus to noise ratio. Following similar results in [DKL⁺23, KLSS23], in Lemmas 8 and 9 we show that these pairs of distributions are instead computationally indistinguishable under the Leaky LWE assumption. Consequently, under appropriate parameters, the desired indistinguishability holds under the LWE assumption.

Lemma 8. *Let params^* , params and ϵ be as specified in Example 2. If the $\text{LWE}_{\text{params}^*}$ assumption holds, then for any $\mathbf{L} \in \mathcal{L}$ it holds that*

$$\left\{ (\mathbf{A}, \mathbf{b}_0, \mathbf{b}_1) \left| \begin{array}{l} \mathbf{A} \leftarrow \mathcal{R}_q^{n \times m} \\ \mathbf{s} \leftarrow \mathcal{R}_q^n; \mathbf{e} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sigma} \\ \mathbf{f} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sigma_f} \\ \mathbf{b}_0^\top := \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \bmod q \\ \mathbf{b}_1^\top := \mathbf{s}^\top \mathbf{A} \mathbf{L} - \mathbf{f}^\top \bmod q \end{array} \right. \right\} \approx_c \left\{ (\mathbf{A}, \mathbf{b}_0, \mathbf{b}_1) \left| \begin{array}{l} \mathbf{A} \leftarrow \mathcal{R}_q^{n \times m} \\ \mathbf{e} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sigma} \\ \mathbf{f} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sigma_f} \\ \mathbf{l}^\top = \mathbf{e}^\top \mathbf{L} + \mathbf{f}^\top \\ \mathbf{b}_0 \leftarrow \mathcal{R}_q^m \\ \mathbf{b}_1^\top := \mathbf{b}_0^\top \mathbf{L} - \mathbf{l}^\top \bmod q \end{array} \right. \right\}.$$

Proof. By Theorem 3, it suffices to show that the two stated distributions are indistinguishable under the $\text{LLWE}_{\text{params}}$ assumption. Suppose towards contradiction that $\mathcal{A}$ is a PPT algorithm which has non-negligible advantage in distinguishing the two stated distributions. Let $\mathcal{B}$ be a PPT algorithm against the $\text{LLWE}_{\text{params}}$ problem. It proceeds as follows: Receive $\mathbf{A}$ from the LLWE challenger, pass $\mathbf{L}$ to the LLWE challenger and receive $(\mathbf{A}, \mathbf{b}, \mathbf{l})$. Let $\mathbf{b}_0 := \mathbf{b}$ and $\mathbf{b}_1 := \mathbf{b}_0^\top \cdot \mathbf{L} - \mathbf{l}^\top \bmod q$. Pass $(\mathbf{A}, \mathbf{L}, \mathbf{b}_0, \mathbf{b}_1)$ to $\mathcal{A}$ and return whatever $\mathcal{A}$ returns.

By construction, the matrix $\mathbf{L}$ always satisfies the admissibility constraint in the LLWE experiment, i.e. $\mathbf{L} \in \mathcal{L}$. We analyse the output of $\mathcal{A}$ to $\mathcal{B}$. Suppose the input to $\mathcal{B}$ is from the $\text{Exp-LLWE}_{\text{params}, \mathcal{B}}^0$ experiment, so that $\mathbf{b}_0 = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \bmod q$ and $\mathbf{l}^\top = \mathbf{e}^\top \mathbf{L} + \mathbf{f}^\top$ for $\mathbf{e} \leftarrow \mathcal{D}_{\mathcal{R}^m, \sigma_e}$ and $\mathbf{f} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sigma_f}$. We have $\mathbf{b}_1^\top = (\mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top) \cdot \mathbf{L} - (\mathbf{e}^\top \mathbf{L} + \mathbf{f}^\top) = \mathbf{s}^\top \mathbf{A} \mathbf{L} - \mathbf{f}^\top \bmod q$ as in the LHS distribution in the lemma statement.

Else, suppose the input to $\mathcal{B}$ is from the $\text{Exp-LLWE}_{\text{params}, \mathcal{B}}^1$ experiment, so that $\mathbf{b}_0 \leftarrow \mathcal{R}_q^m$ is uniformly random and $\mathbf{l}^\top = \mathbf{e}^\top \mathbf{L} + \mathbf{f}^\top$ for $\mathbf{e} \leftarrow \mathcal{D}_{\mathcal{R}^m, \sigma_e}$ and $\mathbf{f} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sigma_f}$. We have $\mathbf{b}_1^\top = \mathbf{b}_0^\top \cdot \mathbf{L} - \mathbf{l}^\top \bmod q$ as in the RHS distribution in the lemma statement.

We conclude that the advantage of $\mathcal{B}$ against the $\text{LLWE}_{\text{params}}$ problem is identical to that of $\mathcal{A}$ against the two concerned distributions, which is non-negligible, a contradiction. This completes the proof. $\square$

Lemma 9 differs slightly from Lemma 8 in that $\mathbf{b}_0$ is withheld from the adversary. It captures another (similar but different) typical setting appearing in security proofs.

Lemma 9. *Let params^* , params and ϵ be as specified in Example 2. If the $\text{LWE}_{\text{params}^*}$ assumption holds, then for any $\mathbf{L} \in \mathcal{L}$ it holds that*

$$\left\{ (\mathbf{A}, \mathbf{b}_1) \left| \begin{array}{l} \mathbf{A} \leftarrow \mathcal{R}_q^{n \times m} \\ \mathbf{s} \leftarrow \mathcal{R}_q^n; \mathbf{f} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sigma_f} \\ \mathbf{b}_1^\top := \mathbf{s}^\top \mathbf{A} \mathbf{L} - \mathbf{f}^\top \bmod q \end{array} \right. \right\} \approx_c \left\{ (\mathbf{A}, \mathbf{b}_1) \left| \begin{array}{l} \mathbf{A} \leftarrow \mathcal{R}_q^{n \times m} \\ \mathbf{b}_0 \leftarrow \mathcal{R}_q^m; \mathbf{f} \leftarrow \mathcal{D}_{\mathcal{R}^Q, \sigma_f} \\ \mathbf{b}_1^\top := \mathbf{b}_0^\top \mathbf{L} - \mathbf{f}^\top \bmod q \end{array} \right. \right\}.$$

Proof. The proof is almost identical to that of Lemma 8, except for a minor difference: In the case where the input to $\mathcal{B}$ is from the $\text{Exp-LLWE}_{\text{params}, \mathcal{B}}^1$ experiment, we can argue that since $\mathbf{b}_0$ is uniformly random and hidden from $\mathcal{A}$ the value $\mathbf{b}_0 - \mathbf{e}$ is also uniformly random. Therefore, $\mathbf{b}_0^\top \mathbf{L} - \mathbf{l}^\top = \mathbf{b}_0^\top \mathbf{L} - \mathbf{e}^\top \mathbf{L} - \mathbf{f}^\top \bmod q$ and $\mathbf{b}_0^\top \mathbf{L} - \mathbf{f}^\top \bmod q$ are identically distributed when given $\mathbf{A}$. $\square$

Game hops resembling Lemma 8 is common in advanced encryption under the dual-Regev paradigm (e.g. [DKL⁺23]), and game hops resembling Lemma 9 is a general setting common in lattice-based zero-knowledge proofs. As shown above, Leaky LWE allows to argue these game hops in a computational manner without resorting to statistical noise-flooding. Eventually, Leaky LWE leads to smaller parameters, thus improved efficiency for practical instantiations of primitives. For concreteness, suppose an application relies on LWE with Q leakage on the LWE secret/error. Using noise-flooding imposes a super-polynomial lower bound on the modulus q , say 2^{128} in practice. Using Leaky LWE and setting parameters against the LWE instance reducible to Leaky LWE (Theorem 3), q can be only polynomially larger than $\sqrt{Q}$. For threshold signatures where Q is the number of signing queries, in practice one may limit Q by $\approx 2^{64}$, meaning q only needs to be slightly larger than 2^{32} . Since larger q also leads to larger LWE secret dimension, such a difference can be decisive for deployability.

5 Conclusion and Discussions

In this work, we have presented the Leaky LWE assumption, a generalisation and unification of the recently proposed Error-Leakage LWE and Hint-MLWE assumptions. Leaky LWE asserts the pseudorandomness of LWE samples in presence of noisy linear leakages on both the LWE secret $\mathbf{s}$ and error $\mathbf{e}$, where the leakage matrix $\mathbf{L}$ can be semi-adaptively chosen by an adversary after seeing the public LWE matrix $\mathbf{A}$. When the LWE secret $\mathbf{s}$ and/or error $\mathbf{e}$ follow Gaussian distributions, we obtained a tight reduction from LWE to Leaky LWE. The Leaky LWE assumption is intended to be a general tool useful for practical instantiations of lattice-based primitives, offering a way to avoid statistical noise-flooding in security proofs. In particular, Lemmas 8 and 9 abstract out technical steps in security proofs of dual-Regev style encryption schemes and lattice-based zero-knowledge proofs, respectively.

Our tight reduction from LWE to Leaky LWE in Theorem 3 relies on Theorems 1 and 2, the latter make use of the convenient properties that Gaussian distributions admit, primarily, that linear transforms of Gaussians remain Gaussians. An interesting extension would be to consider Leaky LWE with non-Gaussian secret and/or error, e.g. bounded uniform distributions or centred binomial distributions, which would be useful for efficient instantiations of lattice-based primitives. However, since the resulting leakage distributions are hard to characterise, obtaining a tight reduction likely requires different techniques.

Another potential generalisation is to consider non-linear leakages, for which we would require non-linear generalisations of Lemmas 4 and 5: For a fixed non-linear function f , we need to explicitly characterise the distribution induced by applying f on Gaussian vectors, which is a challenging task on its own.

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References

- [ACL⁺22] Martin R. Albrecht, Valerio Cini, Russell W. F. Lai, Giulio Malavolta, and Sri Aravinda Krishnan Thyagarajan. Lattice-based SNARKs: Publicly verifiable, preprocessing, and recursively composable - (extended abstract). In Yevgeniy Dodis and Thomas Shrimpton, editors, *CRYPTO 2022, Part II*, volume 13508 of *LNCS*, pages 102–132. Springer, Cham, August 2022. doi:[10.1007/978-3-031-15979-4_4](https://doi.org/10.1007/978-3-031-15979-4_4).
- [ALS16] Shweta Agrawal, Benoît Libert, and Damien Stehlé. Fully secure functional encryption for inner products, from standard assumptions. In Matthew Robshaw and Jonathan Katz, editors, *CRYPTO 2016, Part III*, volume 9816 of *LNCS*, pages 333–362. Springer, Berlin, Heidelberg, August 2016. doi:[10.1007/978-3-662-53015-3_12](https://doi.org/10.1007/978-3-662-53015-3_12).
- [AP12] Jacob Alperin-Sheriff and Chris Peikert. Circular and KDM security for identity-based encryption. In Marc Fischlin, Johannes Buchmann, and Mark Manulis, editors, *PKC 2012*, volume 7293 of *LNCS*, pages 334–352. Springer, Berlin, Heidelberg, May 2012. doi:[10.1007/978-3-642-30057-8_20](https://doi.org/10.1007/978-3-642-30057-8_20).
- [ARS24] Damiano Abram, Lawrence Roy, and Peter Scholl. Succinct homomorphic secret sharing. In Marc Joye and Gregor Leander, editors, *EUROCRYPT 2024, Part VI*, volume 14656 of *LNCS*, pages 301–330. Springer, Cham, May 2024. doi:[10.1007/978-3-031-58751-1_11](https://doi.org/10.1007/978-3-031-58751-1_11).
- [BGV14] Zvika Brakerski, Craig Gentry, and Vinod Vaikuntanathan. (leveled) fully homomorphic encryption without bootstrapping. *ACM Trans. Comput. Theory*, 6(3), July 2014. doi:[10.1145/2633600](https://doi.org/10.1145/2633600).
- [BJTW25] Katharina Boudgoust, Corentin Jeudy, Erkan Tairi, and Weiqiang Wen. Hardness of m-LWE with general distributions and applications to leaky variants. Cryptology ePrint Archive, Paper 2025/1472, 2025. URL: <https://eprint.iacr.org/2025/1472>.
- [BKL⁺25] Cecilia Boschini, Darya Kaviani, Russell WF Lai, Giulio Malavolta, Akira Takahashi, and Mehdi Tibouchi. Ringtail: practical two-round threshold signatures from learning with errors. In *2025 IEEE Symposium on Security and Privacy (SP)*, pages 149–164. IEEE, 2025. doi:[10.1109/SP61157.2025.00070](https://doi.org/10.1109/SP61157.2025.00070).
- [CLM23] Valerio Cini, Russell W. F. Lai, and Giulio Malavolta. Lattice-based succinct arguments from vanishing polynomials - (extended abstract). In Helena Handschuh and Anna Lysyanskaya, editors, *CRYPTO 2023, Part II*, volume 14082 of *LNCS*, pages 72–105. Springer, Cham, August 2023. doi:[10.1007/978-3-031-38545-2_3](https://doi.org/10.1007/978-3-031-38545-2_3).
- [DKL⁺23] Nico Döttling, Dimitris Kolonelos, Russell W. F. Lai, Chuanwei Lin, Giulio Malavolta, and Ahmadreza Rahimi. Efficient laconic cryptography from learning with errors. In Carmit Hazay and Martijn Stam, editors, *EUROCRYPT 2023, Part III*, volume 14006 of *LNCS*, pages 417–446. Springer, Cham, April 2023. doi:[10.1007/978-3-031-30620-4_14](https://doi.org/10.1007/978-3-031-30620-4_14).
- [DKLW25] Adrien Dubois, Michael Klooß, Russell W. F. Lai, and Ivy K. Y. Woo. Lattice-based proof-friendly signatures from vanishing short integer solutions. In Tibor Jager and Jiaxin Pan, editors, *PKC 2025, Part I*, volume 15674 of *LNCS*, pages 452–486. Springer, Cham, May 2025. doi:[10.1007/978-3-031-91820-9_15](https://doi.org/10.1007/978-3-031-91820-9_15).

- [DKM⁺24] Rafaël Del Pino, Shuichi Katsumata, Mary Maller, Fabrice Mouhartem, Thomas Prest, and Markku-Juhani O. Saarinen. Threshold raccoon: Practical threshold signatures from standard lattice assumptions. In Marc Joye and Gregor Leander, editors, *EUROCRYPT 2024, Part II*, volume 14652 of *LNCS*, pages 219–248. Springer, Cham, May 2024. doi:[10.1007/978-3-031-58723-8_8](https://doi.org/10.1007/978-3-031-58723-8_8).
- [GMPW20] Nicholas Genise, Daniele Micciancio, Chris Peikert, and Michael Walter. Improved discrete gaussian and subgaussian analysis for lattice cryptography. In Aggelos Kiayias, Markulf Kohlweiss, Petros Wallden, and Vassilis Zikas, editors, *PKC 2020, Part I*, volume 12110 of *LNCS*, pages 623–651. Springer, Cham, May 2020. doi:[10.1007/978-3-030-45374-9_21](https://doi.org/10.1007/978-3-030-45374-9_21).
- [HMT11] Nathan Halko, Per-Gunnar Martinsson, and Joel A Tropp. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM review*, 53(2):217–288, 2011. doi:[10.1137/090771806](https://doi.org/10.1137/090771806).
- [KLSS23] Duhyeong Kim, Dongwon Lee, Jinyeong Seo, and Yongsoo Song. Toward practical lattice-based proof of knowledge from hint-MLWE. In Helena Handschuh and Anna Lysyanskaya, editors, *CRYPTO 2023, Part V*, volume 14085 of *LNCS*, pages 549–580. Springer, Cham, August 2023. doi:[10.1007/978-3-031-38554-4_18](https://doi.org/10.1007/978-3-031-38554-4_18).
- [LPR10] Vadim Lyubashevsky, Chris Peikert, and Oded Regev. On ideal lattices and learning with errors over rings. In Henri Gilbert, editor, *EUROCRYPT 2010*, volume 6110 of *LNCS*, pages 1–23. Springer, Berlin, Heidelberg, May / June 2010. doi:[10.1007/978-3-642-13190-5_1](https://doi.org/10.1007/978-3-642-13190-5_1).
- [LPR13] Vadim Lyubashevsky, Chris Peikert, and Oded Regev. A toolkit for ring-LWE cryptography. In Thomas Johansson and Phong Q. Nguyen, editors, *EUROCRYPT 2013*, volume 7881 of *LNCS*, pages 35–54. Springer, Berlin, Heidelberg, May 2013. doi:[10.1007/978-3-642-38348-9_3](https://doi.org/10.1007/978-3-642-38348-9_3).
- [LPSS14] San Ling, Duong Hieu Phan, Damien Stehlé, and Ron Steinfeld. Hardness of k-LWE and applications in traitor tracing. In Juan A. Garay and Rosario Gennaro, editors, *CRYPTO 2014, Part I*, volume 8616 of *LNCS*, pages 315–334. Springer, Berlin, Heidelberg, August 2014. doi:[10.1007/978-3-662-44371-2_18](https://doi.org/10.1007/978-3-662-44371-2_18).
- [LS15] Adeline Langlois and Damien Stehlé. Worst-case to average-case reductions for module lattices. *DCC*, 75(3):565–599, 2015. doi:[10.1007/s10623-014-9938-4](https://doi.org/10.1007/s10623-014-9938-4).
- [MP12] Daniele Micciancio and Chris Peikert. Trapdoors for lattices: Simpler, tighter, faster, smaller. In David Pointcheval and Thomas Johansson, editors, *EUROCRYPT 2012*, volume 7237 of *LNCS*, pages 700–718. Springer, Berlin, Heidelberg, April 2012. doi:[10.1007/978-3-642-29011-4_41](https://doi.org/10.1007/978-3-642-29011-4_41).
- [MS24] Daniele Micciancio and Adam Suhl. Simulation-secure threshold PKE from LWE with polynomial modulus. *CiC*, 1(4):2, 2024. doi:[10.62056/a0zogy4e-](https://doi.org/10.62056/a0zogy4e-).
- [Reg05] Oded Regev. On lattices, learning with errors, random linear codes, and cryptography. In Harold N. Gabow and Ronald Fagin, editors, *37th ACM STOC*, pages 84–93. ACM Press, May 2005. doi:[10.1145/1060590.1060603](https://doi.org/10.1145/1060590.1060603).

- [SSTX09] Damien Stehlé, Ron Steinfeld, Keisuke Tanaka, and Keita Xagawa. Efficient public key encryption based on ideal lattices. In Mitsuru Matsui, editor, *ASIACRYPT 2009*, volume 5912 of *LNCS*, pages 617–635. Springer, Berlin, Heidelberg, December 2009. doi:10.1007/978-3-642-10366-7_36.
- [Wee22] Hoeteck Wee. Optimal broadcast encryption and CP-ABE from evasive lattice assumptions. In Orr Dunkelman and Stefan Dziembowski, editors, *EUROCRYPT 2022, Part II*, volume 13276 of *LNCS*, pages 217–241. Springer, Cham, May / June 2022. doi:10.1007/978-3-031-07085-3_8.
- [Wee24] Hoeteck Wee. Circuit ABE with $\text{poly}(\text{depth}, \lambda)$ -sized ciphertexts and keys from lattices. In Leonid Reyzin and Douglas Stebila, editors, *CRYPTO 2024, Part III*, volume 14922 of *LNCS*, pages 178–209. Springer, Cham, August 2024. doi:10.1007/978-3-031-68382-4_6.
- [WLL24] Zhedong Wang, Qiqi Lai, and Feng-Hao Liu. Ring/module learning with errors under linear leakage - hardness and applications. In Qiang Tang and Vanessa Teague, editors, *PKC 2024, Part II*, volume 14602 of *LNCS*, pages 275–304. Springer, Cham, April 2024. doi:10.1007/978-3-031-57722-2_9.

A Discussion on Theorems 2 and 3 of [DKL⁺23]

In the work of [DKL⁺23], the authors provided [DKL⁺23, Theorem 2] which is conceptually similar to our Theorem 2, with the main difference being that both its statement and proof additionally require the use of continuous Gaussian distributions. Similar to our Theorem 3, the authors then used [DKL⁺23, Theorem 2] to construct a reduction from standard LWE to Error-Leakage LWE, summarised by [DKL⁺23, Theorem 3]. We discuss some discrepancies around these two theorems and their proofs. We adopt their notations below for ease of references.

Proof of [DKL⁺23, Theorem 2]. A major argument in the proof of [DKL⁺23, Theorem 2] is to lower-bound the singular values of a matrix $\mathbf{S}\mathbf{S}^T$, where $\mathbf{S} = \mathbf{Z}\sqrt{\Sigma_2}$, with $\mathbf{Z}$ denotes the transpose of the leakage matrix (i.e. $\mathbf{L}^T$ in the notation of this work). This argument appears in the proof of [DKL⁺23, Lemma 8] (which is required in the proof of [DKL⁺23, Theorem 2]). To do so, the authors in turn relied on [DKL⁺23, Lemmas 6 and 7]. However, a subtle yet important requirement from the latter two lemmas is that $\mathbf{S}\mathbf{S}^T$ must be a full-rank matrix (a necessary condition for the provided proofs to go through). This translates to requiring the leakage matrix $\mathbf{Z}^T$ to be square and full-rank, a restriction that is unlikely satisfied in the context of [DKL⁺23, Lemma 8, Theorem 2]. Indeed, such restricted class of leakage matrices is insufficient for proving the Error-Leakage LWE assumption as stated, nor be used in their RBE application. We provide Lemma 10, which can be seen as adaption of [DKL⁺23, Lemmas 6 and 7] to the setting required in the proof. Lemma 10 can be plugged into their proof to close the above gap.

Lemma 10. *Let $m, n \in \mathbb{N}$ be arbitrary, and let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be any matrix. It holds that*

$$\mathbf{A}^T(\mathbf{I}_m + \mathbf{A}\mathbf{A}^T)^{-1}\mathbf{A} \leq (1 + s_{\max}(\mathbf{A}\mathbf{A}^T)^{-1})^{-1}.$$

Proof. Below we proof the claim for the case $m \geq n$, and we sketch the difference for the case $m < n$ at the end.

Let $\mathbf{A} = \mathbf{U} \begin{pmatrix} \mathbf{D} \\ \mathbf{0} \end{pmatrix} \mathbf{V}^T$ be the singular value decomposition of $\mathbf{A}$, where $\mathbf{U} \in \mathbb{R}^{m \times m}$, $\mathbf{V} \in \mathbb{R}^{n \times n}$ are unitary, and $\mathbf{D} \in \mathbb{R}^{n \times n}$ is a diagonal matrix with non-negative entries. We have

$$\mathbf{A}\mathbf{A}^T = \left(\mathbf{U} \begin{pmatrix} \mathbf{D} \\ \mathbf{0} \end{pmatrix} \mathbf{V}^T \right) \cdot (\mathbf{V} (\mathbf{D} \quad \mathbf{0}) \mathbf{U}^T) = \mathbf{U} \begin{pmatrix} \mathbf{D}^2 & \\ & \mathbf{0} \end{pmatrix} \mathbf{U}^T,$$

$$\mathbf{A}^\top \mathbf{A} = (\mathbf{V} (\mathbf{D} \ \mathbf{0}) \mathbf{U}^\top) \cdot \left(\mathbf{U} \begin{pmatrix} \mathbf{D} \\ \mathbf{0} \end{pmatrix} \mathbf{V}^\top \right) = \mathbf{V} \mathbf{D}^2 \mathbf{V}^\top.$$

Therefore

$$\begin{aligned} & \mathbf{A}^\top (\mathbf{I}_m + \mathbf{A} \mathbf{A}^\top)^{-1} \mathbf{A} \\ &= \mathbf{A}^\top \cdot \left(\mathbf{I}_m + \mathbf{U} \begin{pmatrix} \mathbf{D}^2 & \\ & \mathbf{0} \end{pmatrix} \mathbf{U}^\top \right)^{-1} \cdot \mathbf{A} \\ &= \mathbf{A}^\top \cdot \left(\mathbf{U} \cdot \left(\mathbf{I}_m + \begin{pmatrix} \mathbf{D}^2 & \\ & \mathbf{0} \end{pmatrix} \right) \cdot \mathbf{U}^\top \right)^{-1} \cdot \mathbf{A} \\ &= \mathbf{A}^\top \cdot \mathbf{U}^{-\top} \cdot \begin{pmatrix} \mathbf{I}_n + \mathbf{D}^2 & \\ & \mathbf{I}_{m-n} \end{pmatrix}^{-1} \cdot \mathbf{U}^{-1} \cdot \mathbf{A} \\ &= (\mathbf{V} (\mathbf{D} \ \mathbf{0}) \mathbf{U}^\top) \cdot \mathbf{U}^{-\top} \cdot \begin{pmatrix} (\mathbf{I}_n + \mathbf{D}^2)^{-1} & \\ & \mathbf{I}_{m-n} \end{pmatrix} \cdot \mathbf{U}^{-1} \cdot \left(\mathbf{U} \begin{pmatrix} \mathbf{D} \\ \mathbf{0} \end{pmatrix} \mathbf{V}^\top \right) \\ &= \mathbf{V} (\mathbf{D} \ \mathbf{0}) \cdot \begin{pmatrix} (\mathbf{I}_n + \mathbf{D}^2)^{-1} & \\ & \mathbf{I}_{m-n} \end{pmatrix} \cdot \begin{pmatrix} \mathbf{D} \\ \mathbf{0} \end{pmatrix} \mathbf{V}^\top \\ &= \mathbf{V} \cdot \mathbf{D} \cdot (\mathbf{I}_n + \mathbf{D}^2)^{-1} \cdot \mathbf{D} \cdot \mathbf{V}^\top. \end{aligned}$$

Write $\mathbf{D} = \text{diag}((d_i)_{i \in [n]})$, then

$$\begin{aligned} \mathbf{D} \cdot (\mathbf{I}_n + \mathbf{D}^2)^{-1} \cdot \mathbf{D} &= \text{diag} \left(\left(\frac{d_i^2}{1 + d_i^2} \right)_{i \in [n]} \right) \\ &\leq \max_{i \in [n]} (1 + d_i^{-2})^{-1} \\ &= (1 + s_{\max}(\mathbf{A} \mathbf{A}^\top)^{-1})^{-1}. \end{aligned}$$

Since multiplication by orthogonal matrices do not alter singular values, we obtain

$$\mathbf{A}^\top (\mathbf{I}_m + \mathbf{A} \mathbf{A}^\top)^{-1} \mathbf{A} \leq (1 + s_{\max}(\mathbf{A} \mathbf{A}^\top)^{-1})^{-1}.$$

For $m < n$, the proof is analogous. We have instead

$$\begin{aligned} \mathbf{A} \mathbf{A}^\top &= (\mathbf{U} (\mathbf{D} \ \mathbf{0}) \mathbf{V}^\top) \cdot \left(\mathbf{V} \begin{pmatrix} \mathbf{D} \\ \mathbf{0} \end{pmatrix} \mathbf{U}^\top \right) = \mathbf{U} \mathbf{D}^2 \mathbf{U}^\top, \\ \mathbf{A}^\top \mathbf{A} &= \left(\mathbf{V} \begin{pmatrix} \mathbf{D} \\ \mathbf{0} \end{pmatrix} \mathbf{U}^\top \right) \cdot (\mathbf{U} (\mathbf{D} \ \mathbf{0}) \mathbf{V}^\top) = \mathbf{V} \begin{pmatrix} \mathbf{D}^2 & \\ & \mathbf{0} \end{pmatrix} \mathbf{V}^\top, \end{aligned}$$

and with similar derivation we obtain

$$\mathbf{A}^\top (\mathbf{I}_m + \mathbf{A} \mathbf{A}^\top)^{-1} \mathbf{A} = \mathbf{V} \cdot \begin{pmatrix} \mathbf{D} (\mathbf{I}_n + \mathbf{D}^2)^{-1} \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \cdot \mathbf{V}^\top.$$

Again, that $\mathbf{D} (\mathbf{I}_n + \mathbf{D}^2)^{-1} \mathbf{D} \leq (1 + s_{\max}(\mathbf{A} \mathbf{A}^\top)^{-1})^{-1}$ implies the claim. $\square$

Application of [DKL⁺23, Theorem 2] in [DKL⁺23, Theorem 3]. The LWE to Error-Leakage LWE reduction [DKL⁺23, Theorem 3] is almost a direct invocation of [DKL⁺23, Theorem 2]. However, we observe two discrepancies between these two theorems:

- In [DKL⁺23, Theorem 3], the relation between the lattice dimensions and the smoothing parameter requirements is inconsistent with that stated in [DKL⁺23, Theorem 2]. Upon inspection, the theorem should require $\mathbf{I}_m \otimes \Sigma_1 \geq \eta_\epsilon(\mathcal{R}^m)$ and $\mathbf{I}_Q \otimes \Sigma_2 \geq \eta_\epsilon(\mathcal{R}^Q)$, which is a stronger requirement than the stated condition of $\Sigma_1, \Sigma_2 \geq \eta_\epsilon(\mathcal{R})$.

- The parameter β in [DKL⁺23, Theorem 2] denotes an upper bound on $s_{\max}(\mathbf{L})$; however, when used in [DKL⁺23, Theorem 3], this was treated as the infinity norm bound of $\mathbf{L}$. This leads to a gap of $\approx \sqrt{\varphi Q}$ in their norm bound condition on $\mathbf{L}$, where φ is the ring degree and Q is the number of columns of $\mathbf{L}$. In the intended use case where Q is a large polynomial, e.g. $Q = 2^{64}$, this leads to a significant mismatch in parameter choices. Fortunately, despite this issue in [DKL⁺23, Theorem 3], in their subsequently RBE application, such $\sqrt{\varphi Q}$ factor was taken into account.